MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, TTL, LOW POWER, FLIP-FLOPS

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power, bistable logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit
01	R-S master-slave flip-flop
02	J-K master-slave flip-flop
03	Dual J-K master-slave flip-flop
04	Dual J-K master-slave flip-flop
05	Dual D-type edge-triggered flip-flop

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	0 V dc to 8.0 V dc
Input voltage range	0 V dc to 6.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation in accordance with flip-flop (P_D) 1/	11 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}): Cases A, B, C, and D	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short-circuit conditions (e.g., I_{OS}) at one output for 5 seconds duration.
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage(V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.7 V dc, except clock input of types 01, 02, 03, and 04
Maximum low-level clock input voltage	0.6 V dc (types 01, 02, 03, and 04)
Normalized fanout (each output) 3/	10 maximum
Width of clock pulse	$\geq$ 200 ns
Width of preset pulse	$\geq$ 100 ns
Width of clear pulse	$\geq$ 100 ns
Input set-up time:	
Device types 02, 03, and 04	> Clock pulse width, minimum
Device type 01	100 ns, minimum when R, S input data is complementary
Device type 01	$\geq$ Clock pulse width, minimum when R, S input data is not complementary
Device type 05	50 ns, minimum
Input hold time	10 ns, minimum
Case operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3/ The device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type					Limits		Unit
			01	02	03	04	05	Min	Max	
High-level output voltage	V_{OH}	V_{IN} = 0.7 V V_{CC} = 4.5 V I_{OH} = -100 µA						2.4		V
Low-level input voltage	V_{OL}	I_{OL} = 2 mA V_{CC} = 4.5 V							0.3	V
Low-level input current	I_{IL1}	V_{CC} = 5.5 V V_{IN} = 0.3 V	S R	J K	J K	J K		-43	-140	µA
Low-level input current	I_{IL2}	V_{CC} = 5.5 V V_{IN} = 0.3 V	Clock Preset Clear	Clock Preset Clear	Clock Clear	Preset		1/ -86	-360 -280	µA µA
Low-level input current	I_{IL3}	V_{CC} = 5.5 V V_{IN} = 0.3 V				Clock Clear		-172	-560	µA
Low-level input current	I_{IL4}	V_{CC} = 5.5 V V_{IN} = 0.3 V					D Preset	-60	-180	µA
Low-level input current	I_{IL5}	V_{CC} = 5.5 V V_{IN} = 0.3 V					Clock Clear	-120	-360	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V V_{IN} = 2.4 V	S R	J K	J K	J K	D		10	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V V_{IN} = 5.5 V	S R	J K	J K	J K	D		100	µA
High-level input current	I_{IH3}	V_{CC} = 5.5 V V_{IN} = 2.4 V	Clear Preset	Clear Preset	Clear	Preset	Clock Preset		20	µA
High-level input current	I_{IH4}	V_{CC} = 5.5 V V_{IN} = 5.5 V	Clear Preset Clock	Clear Preset Clock	Clock Clear	Preset	Clock Clear		200	µA
High-level input current	I_{IH5}	V_{CC} = 5.5 V V_{IN} = 2.4 V					Clear		30	µA
High-level input current	I_{IH6}	V_{CC} = 5.5 V V_{IN} = 5.5 V					Clear		300	µA
High-level input current	I_{IH7}	V_{CC} = 5.5 V V_{IN} = 2.4 V				Clear			40	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type					Limits		Unit
			01	02	03	04	05	Min	Max	
High-level input current	I_{IH8}	V_{CC} = 5.5 V V_{IN} = 5.5 V				Clock Clear			400	µA
High-level input current	I_{IH9}	V_{CC} = 5.5 V V_{IN} = 2.4 V				Clock		0	-400	µA
High-level input current	I_{IH10}	V_{CC} = 5.5 V V_{IN} = 2.4 V	Clock	Clock	Clock			0	-200	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V V_{IN} = 0 2/						-3	-15	mA
Supply current per flip-flop	I_{CC}	V_{CC} = 5.5 V $V_{IN(clock)}$ = 0	Types 01, 02, 03, and 04						1.90	mA
			Type 05						1.50	mA
Maximum clock frequency	f_{MAX} 3/	C_L = 50 pF R_L = 4 kΩ						2.5		MHz
Propagation delay to a high level (clear or preset to output)	t_{PLH}							10	125	ns
Propagation delay to a low level (clear or preset to output)	t_{PHL}		$V_{IN(clock)}$ = 2.4 V					10	200	ns
Propagation delay to a high level (clock to output)	t_{PLH}		$V_{IN(clock)}$ = 0 V, types 01, 02 03, and 04					10	250	ns
Propagation delay to a low level (clock to output)	t_{PHL}							10	125	ns
								10	200	ns

1/ Not more than one output should be shorted at a time.
2/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one half of the input frequency.
3/ For device types 01 and 03.
I_{IL2} = -120 µA, for device type 02
I_{IL2} = -105 µA.

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3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.3 Truth tables and logic diagrams. The truth tables and logic diagrams shall be as specified on figure 2.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are specified in table I, and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3, 9,10,11	1*,2,3, 9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,8, 9,10,11
Group B test requirements (method 5005)	1,2,3,7,8, 9,10,11	N/A
Group C end point electrical parameters (method 5005)	N/A	1,2,3
Group D end point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.3c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 17 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuits shown on figure 4, or equivalent.

(2) T_A = +125°C, minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuits shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

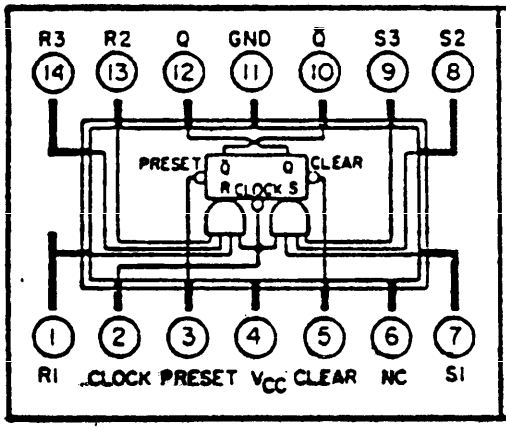
4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

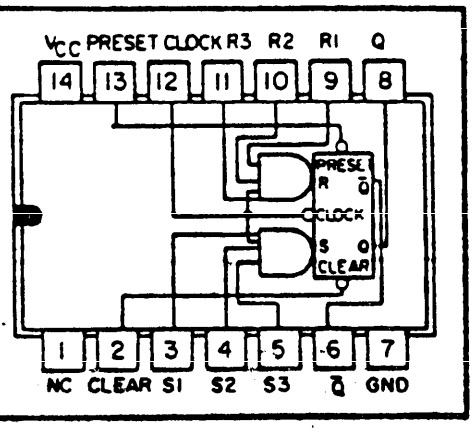
5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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Cases A, B, and D

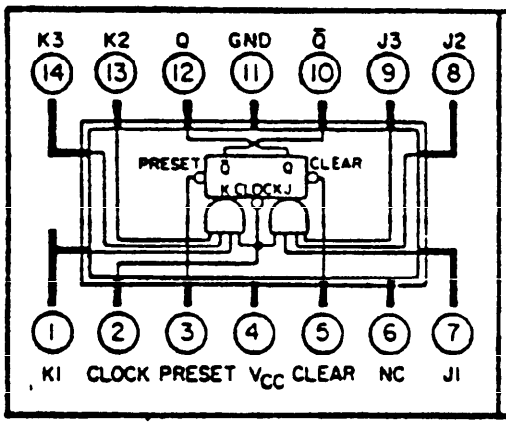


Case C

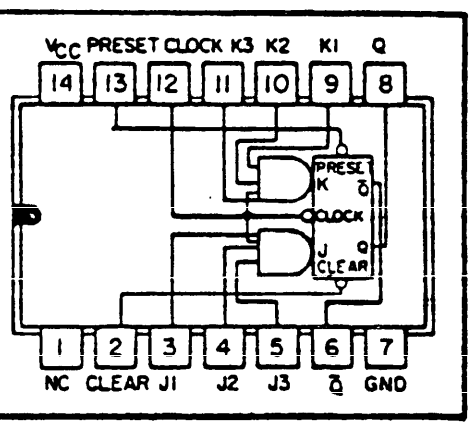


Device type 01

Cases A, B, and D

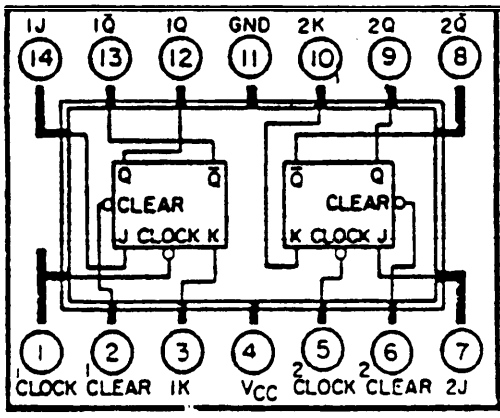


Case C

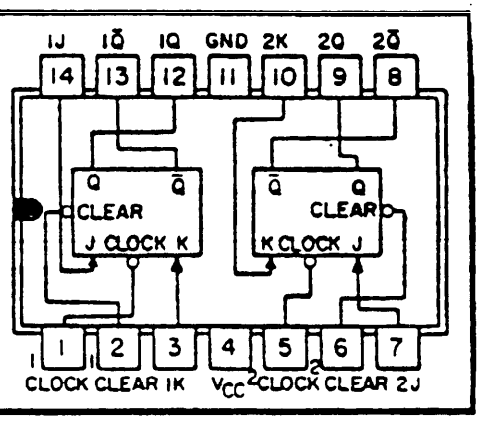


Device type 02

Cases A, B, and D



Case C

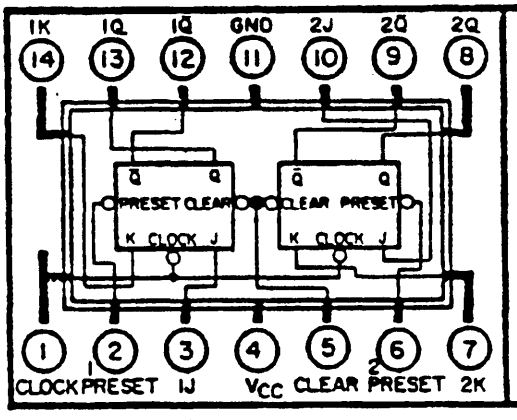


Device type 03

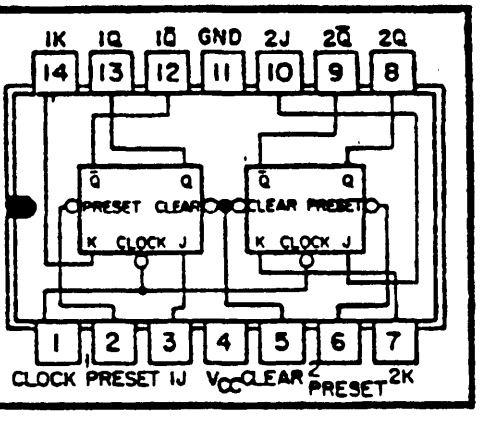
FIGURE 1. Terminal connections.

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Cases A, B, and D

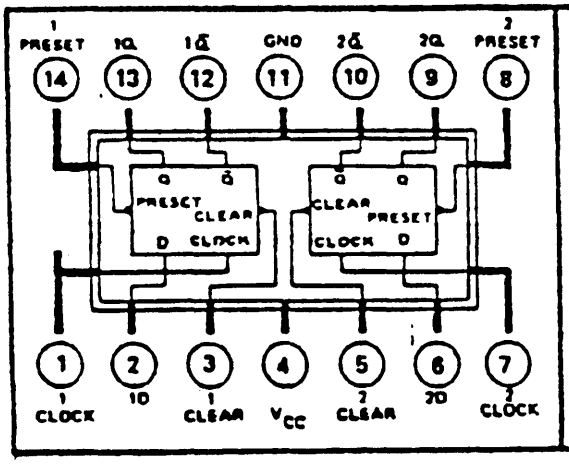


Case C

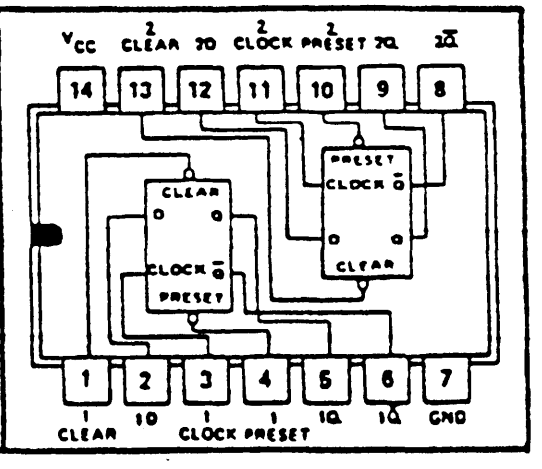


Device type 04

Cases A, B, and D



Case C



Device type 05

FIGURE 1. Terminal connections - Continued.

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Positive logic: Low input to present sets Q to high-level
Low input to clear sets Q to low-level
Preset and clear are independent of clock

Truth table		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

NOTES:
1. J = J1 J2 J3
2. K = K1 K2 K3
3. t_n = Bit time before clock pulse.
4. t_{n+1} = Bit time after clock pulse.

Truth table		
t_n		t_{n+1}
R	S	Q
L	L	Q_n
L	H	H
H	L	L
H	H	Indeterminate

Device type 01.

Positive logic: Low input to present sets Q to high-level
Low input to clear set Q to low-level
Preset and clear are independent of clock

NOTES:
1. R = R1 R2 R3
2. S = S1 S2 S3
3. t_n = Bit time before clock pulse.
4. t_{n+1} = Bit time after clock pulse.

Device type 02.

Description for device types 01 and 02

These flip-flops are based on the master-slave principle and each has AND gate inputs for entry into the master section which are controlled by the clock pulse. The clock pulse also regulates the state of the coupling transistors which connect the master and slave sections. The sequence of operation as controlled by the clock pulse is as follows:

1. Isolate slave from master.
2. Enter information from AND gate inputs to master.
3. Disable AND gate inputs.
4. Transfer information from master to slave.

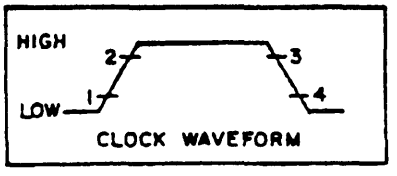


FIGURE 2. Truth tables and device descriptions.

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Truth table each flip-flop		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to present sets Q to high-level
Low input to clear sets Q to low-level
Preset and clear are independent of clock

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

Truth table each flip-flop		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Device type 04

Positive logic: Low input to clear sets Q to low-level
Clear is independent of clock

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

Device type 03

Description for device types 03 and 04

These flip-flops are based on the master-slave principle and each has AND gate inputs for entry into the master section which are controlled by the clock pulse. The clock pulse also regulates the state of the coupling transistors which connect the master and slave sections. The sequence of operation as controlled by the clock pulse is as follows:

1. Isolate slave from master.
2. Enter information from AND gate inputs to master.
3. Disable AND gate inputs.
4. Transfer information from master to slave.

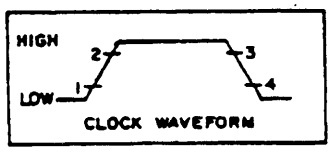


FIGURE 2. Truth tables and device descriptions - Continued.

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Truth table each flip-flop		
t_n	t_{n+1}	
Input D	Output Q	Output $\overline{Q}$
L	L	H
H	H	L

Positive logic: Low input to preset sets Q to high-level
Low input to clear sets Q to low-level
Clear and preset are independent of clock

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

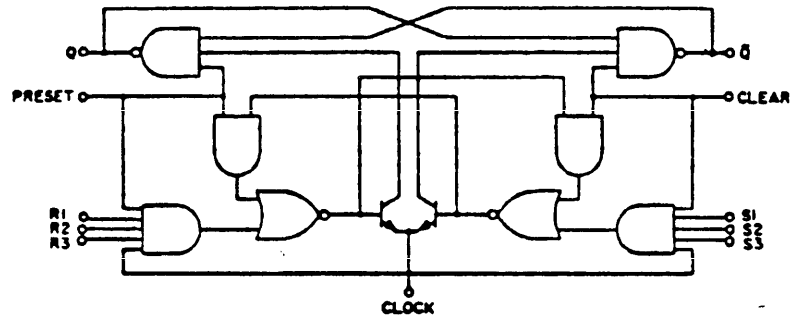
Device type 05

Description for device type 05.

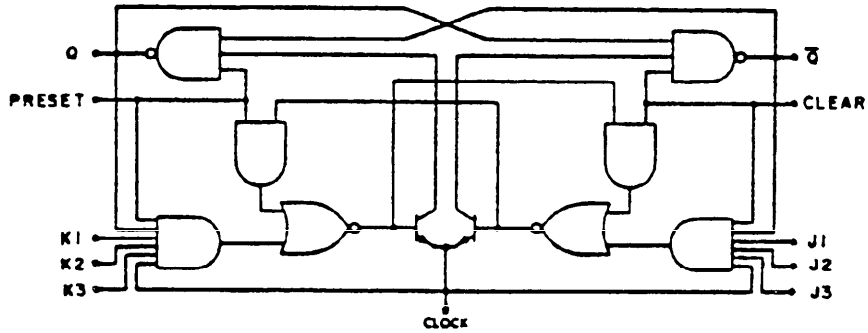
Input information is transferred to the outputs on the positive edge of the clock pulse. Clock triggering occurs at a voltage level of the clock pulse and is not directly related to the transition time of the positive going pulse. After the clock input threshold voltage has been passed, the data input (D) is locked out.

FIGURE 2. Truth tables and device descriptions - Continued.

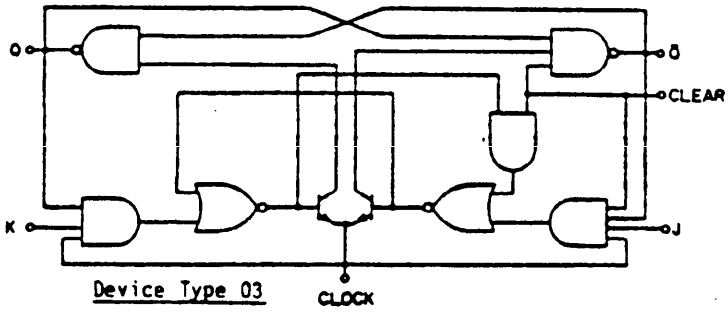
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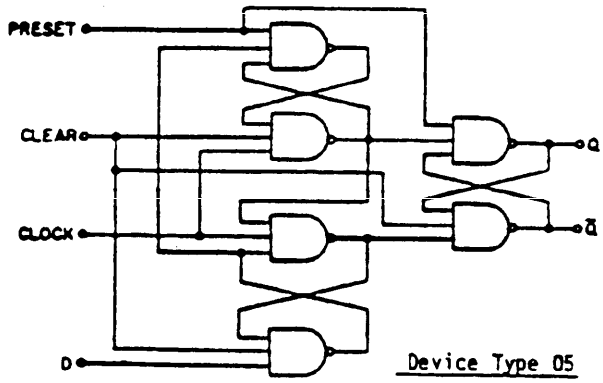
Device Type 01



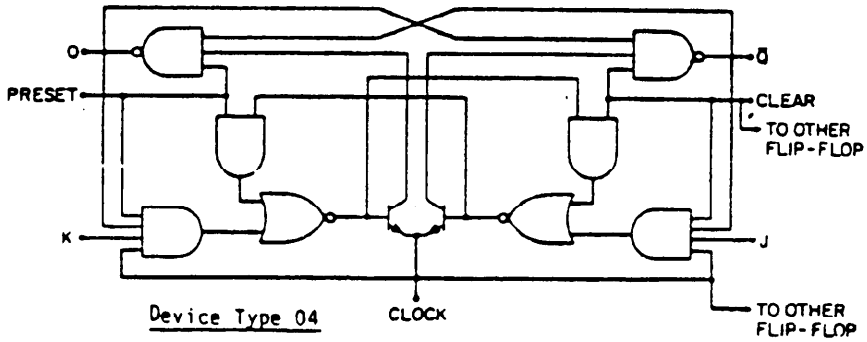
Device Type 02



Device Type 03



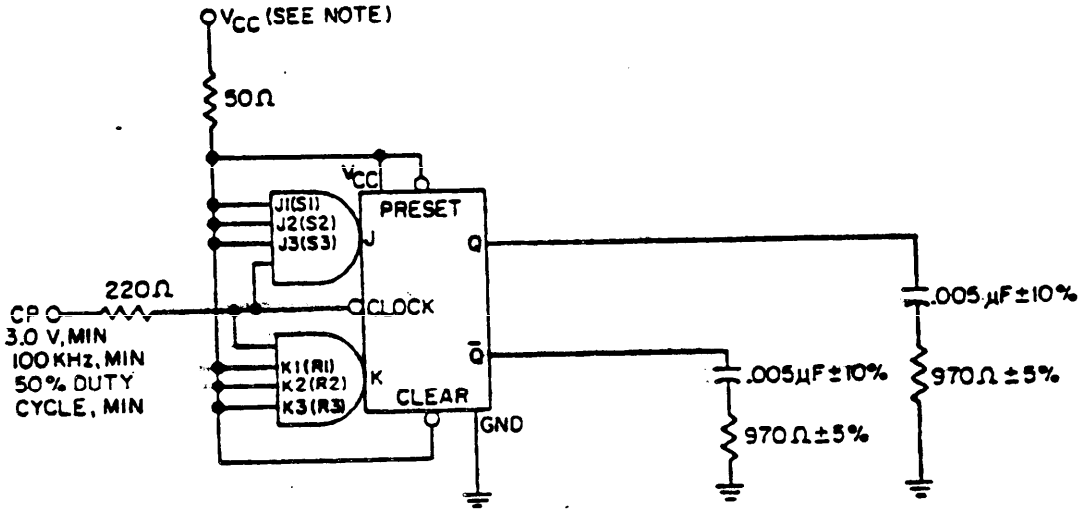
Device Type 05



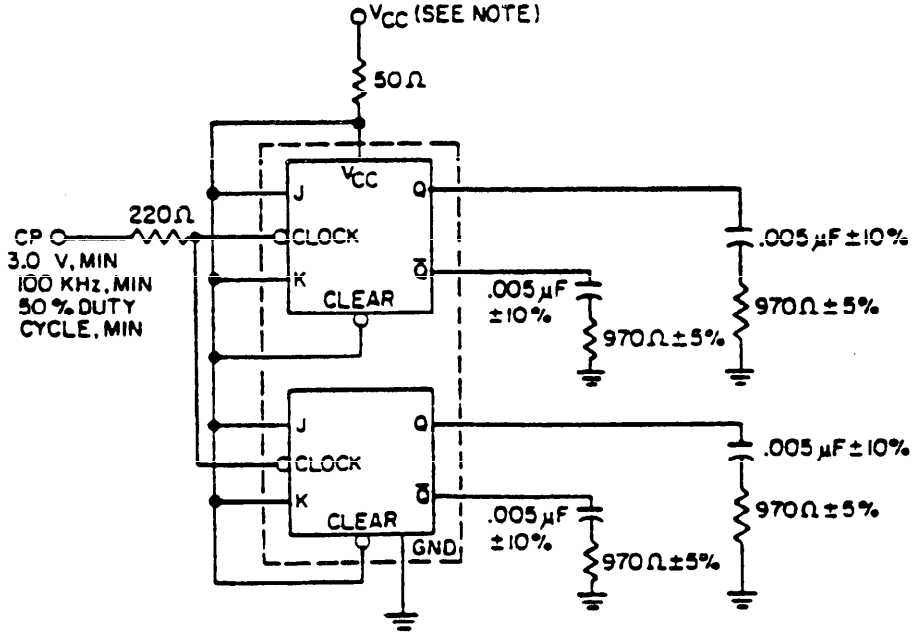
Device Type 04

FIGURE 3. Logic diagram for device types 01, 02, 03, 04, and 05.

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Device types 01 and 02

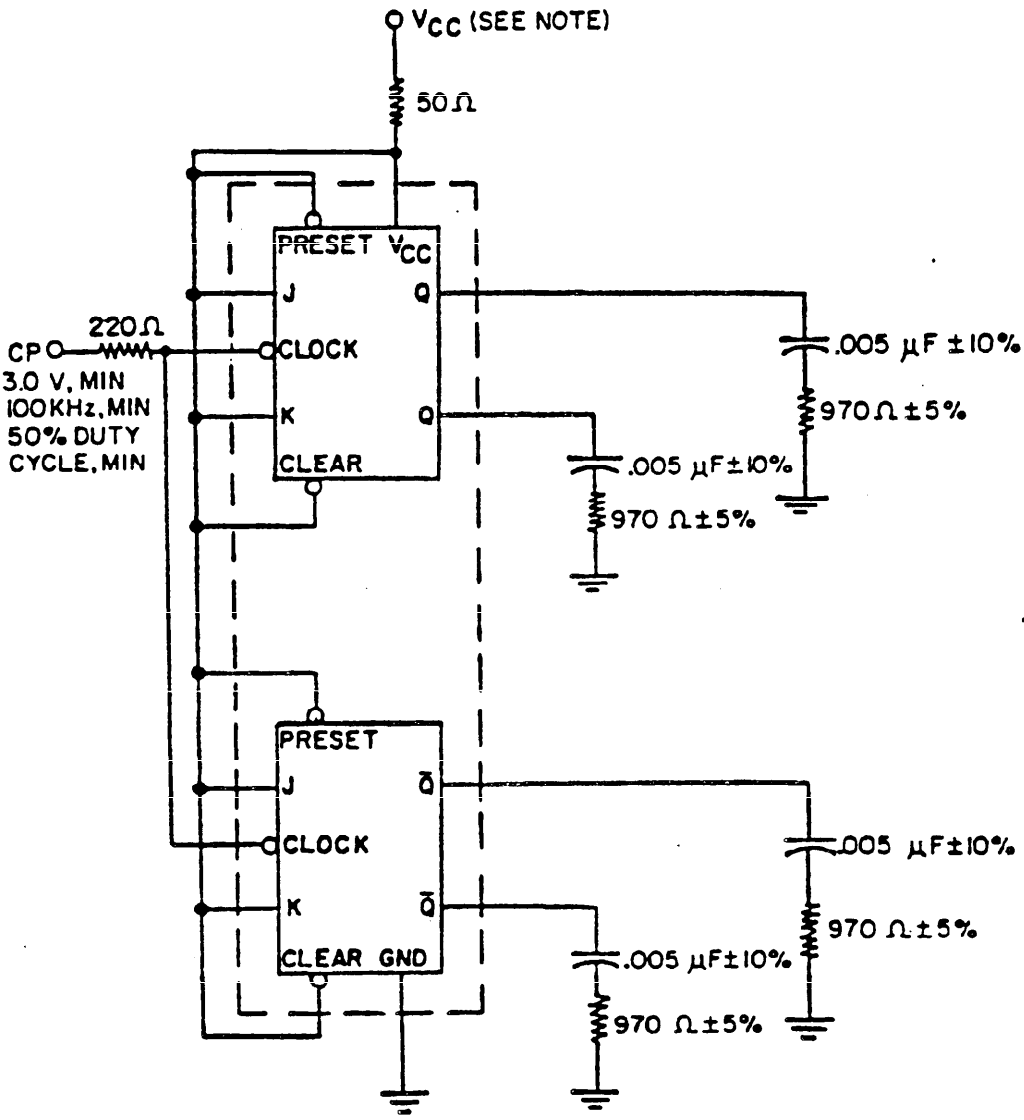


Device type 03

NOTE: V_{CC} = 5 volts, minimum, at the device terminals.

FIGURE 4. Burn-in and life test circuits.

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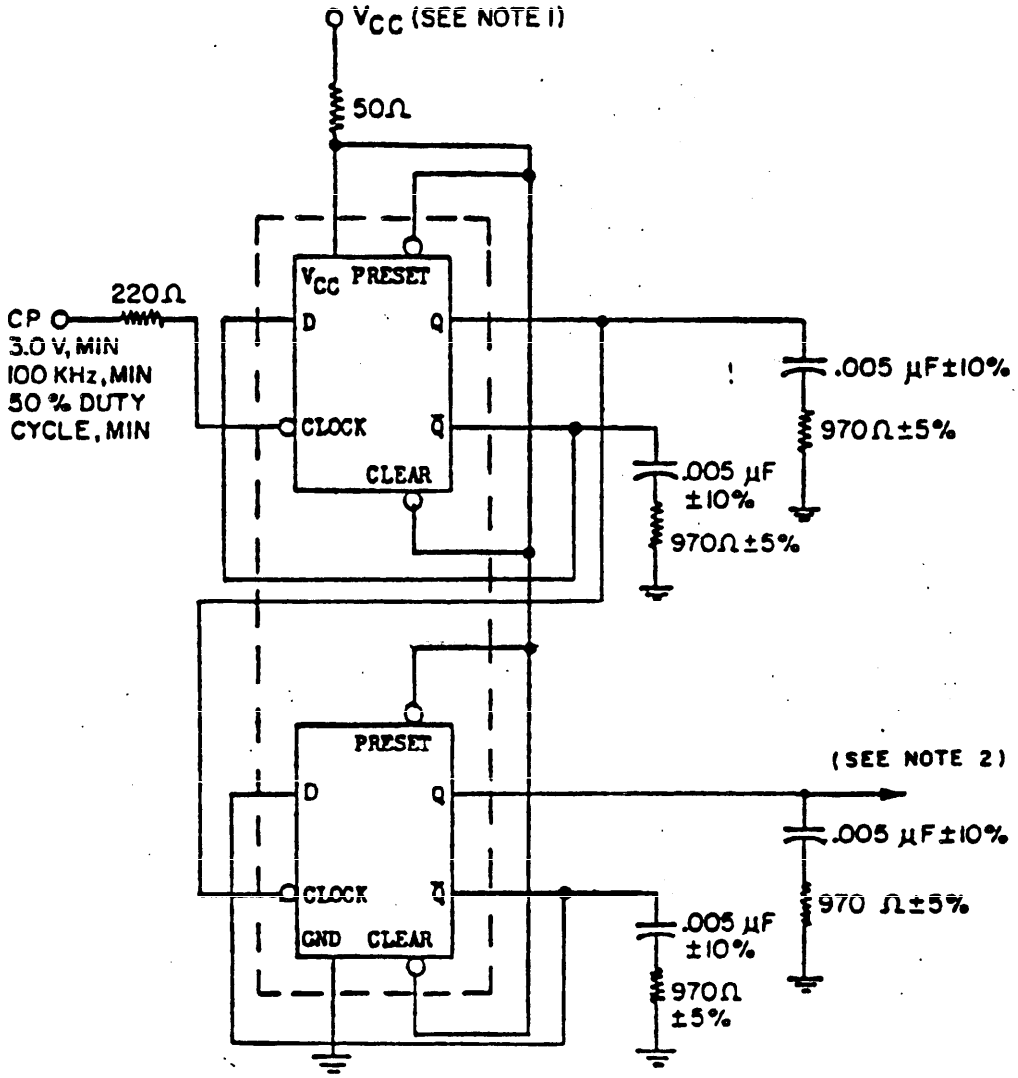


Device type 04

NOTE: V_{CC} = 5 volts, minimum, at the device terminals.

FIGURE 4. Burn-in and life test circuits - Continued.

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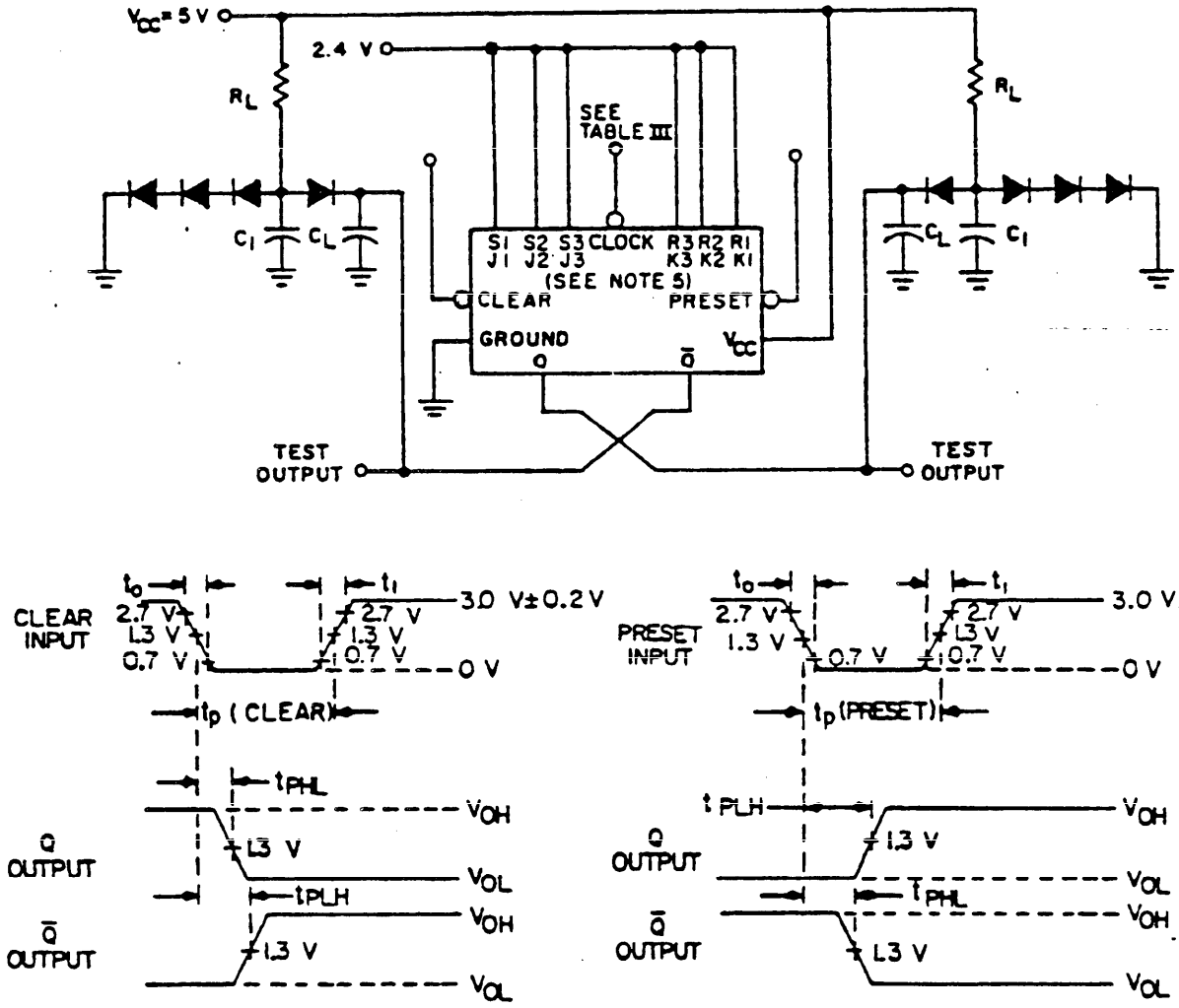


Device Type 05

1. V_{CC} = 5 volts, minimum, at the device terminals.
2. To other devices when testing in series.

FIGURE 4. Burn-in and life test circuits - Continued.

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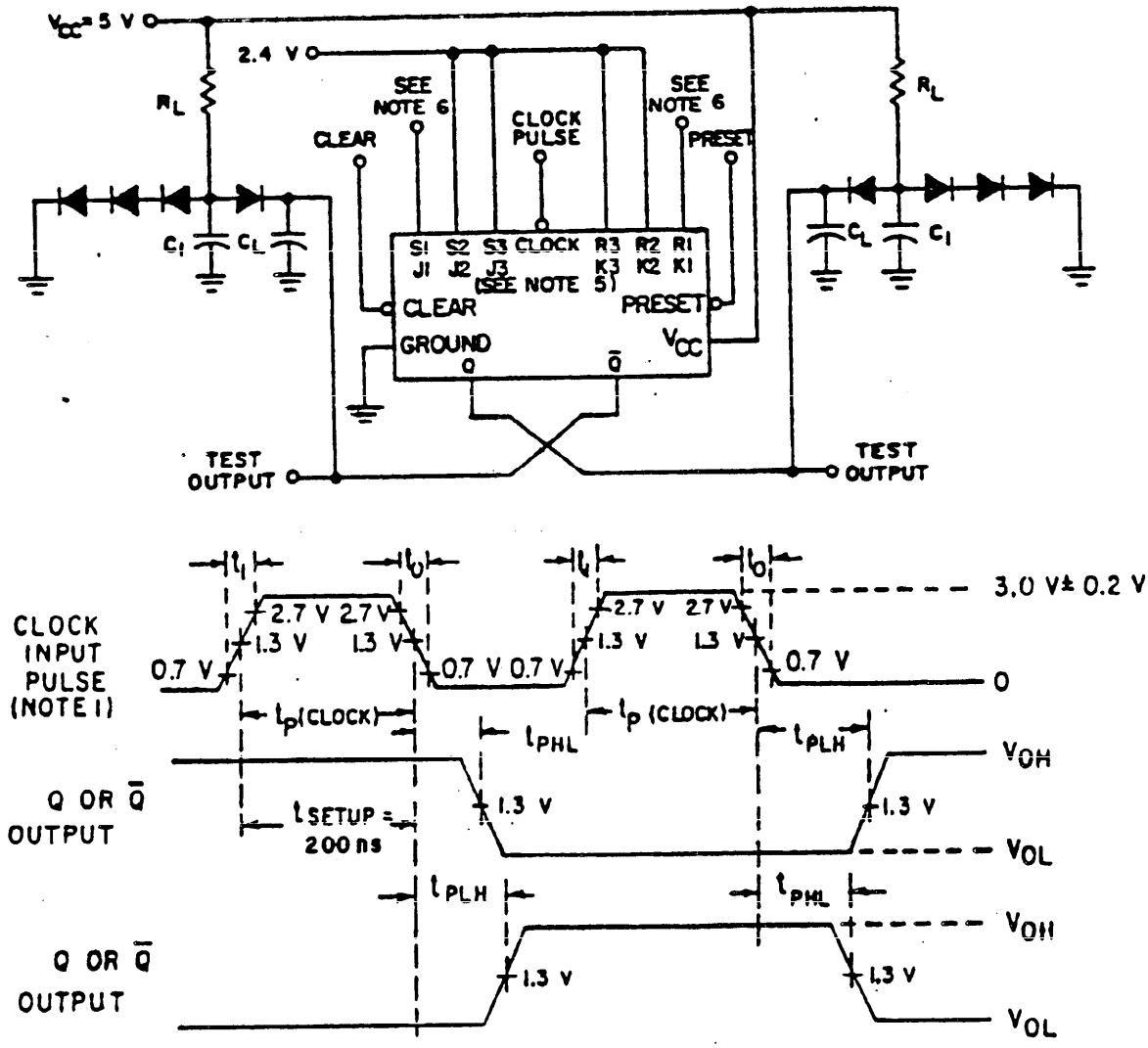


NOTES:

1. Clear or preset input pulse characteristics: V_{gen} = 3.0 V ±0.2 V, t_0 = 15 ns, t_1 = 15 ns, t_p (clear) = t_p (preset) = 100 ns, PRR = 0.5 MHz and Z_{out} = 50 Ω.
2. C_L = 50 pf minimum and includes probe and jig capacitance.
3. R_L = 4 kΩ ±5% and C_l = 30 pF minimum.
4. All diodes are 1N916 or equivalent.
5. R and S inputs apply for device type 01, J and K inputs apply for device type 02.
6. When testing clear to output switching, preset input shall have a negative pulse; when testing preset to output switching, clear input shall have a negative pulse (see Table III).1

FIGURE 5. Clear and preset switching test circuit for device type 01 and 02.

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NOTES:
1. Clock input pulse characteristics: V_{gen} = 3.0 V ±0.2 V,

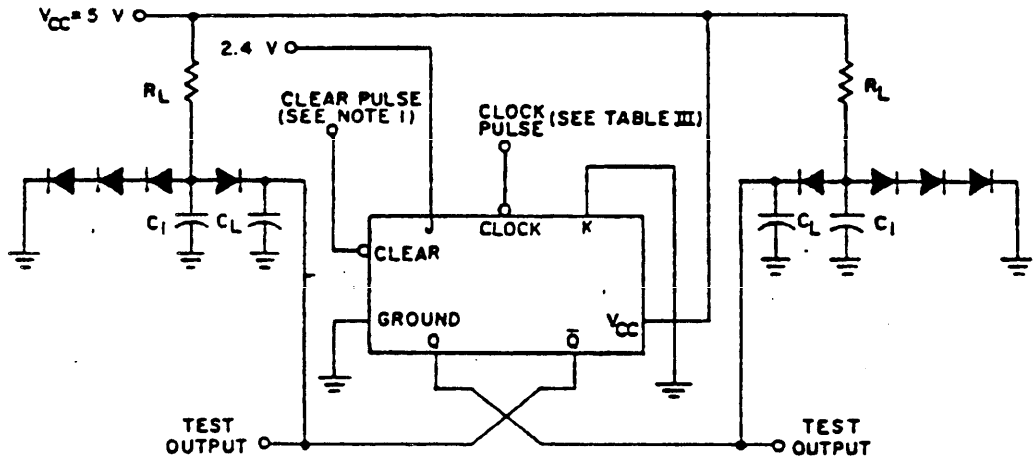
   t_0 = 15 ns, t_1 = 15 ns, t_p = 200 ns, and PRR = 0.5 MHz.

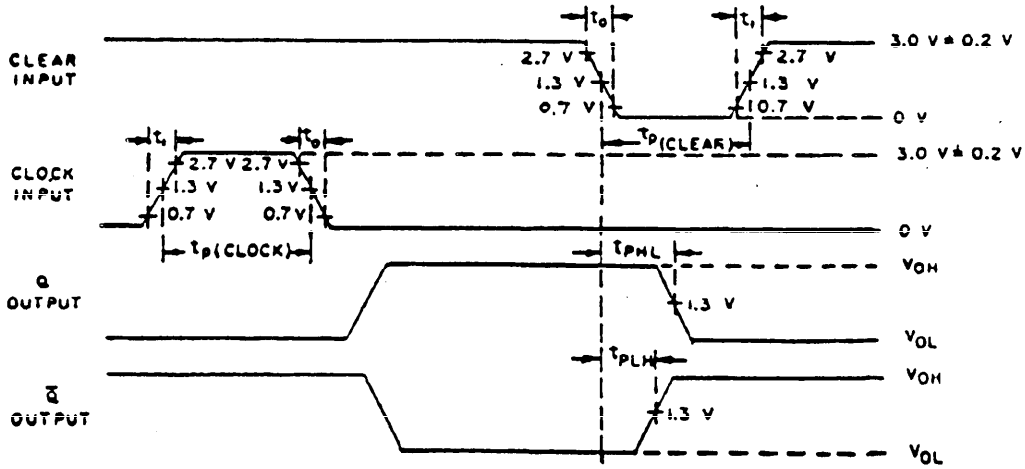
   when testing F_{max}, PRR = see table III.

2. All diodes are 1N916 or equivalent.
3. C_L = 50 pF minimum and includes probe and jig capacitance.
4. R_L = 4kΩ ±5% and C_t = 30 pF minimum.
5. R and S inputs apply for device type 01, J and K inputs apply for device type 02.
6. R1 input is connected to Q output, S1 input is connected to $\bar{Q}$ output. J1 and K1 inputs are connected to 2.4 V.

FIGURE 6. Synchronous switching test circuit for devices types 01 and 02.

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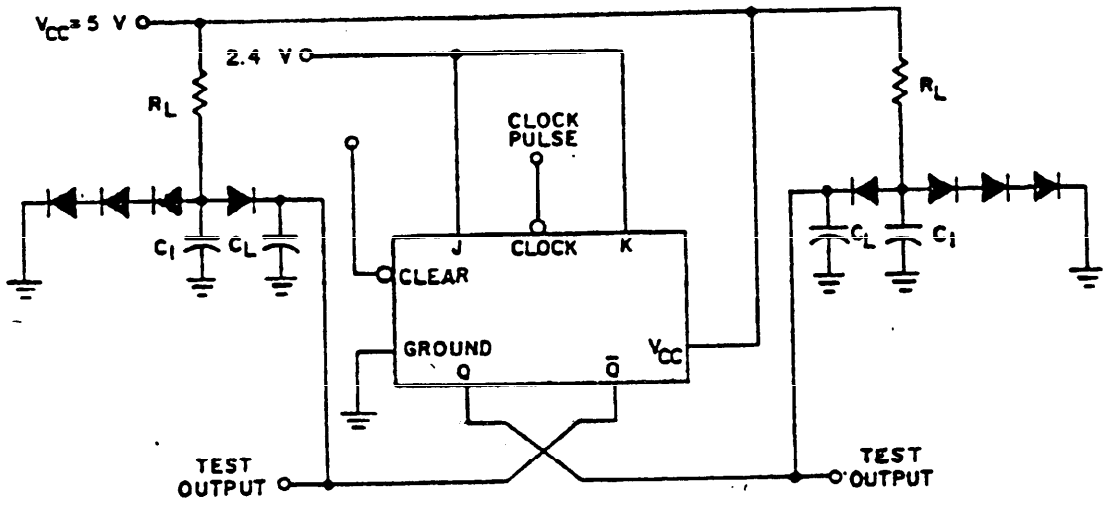


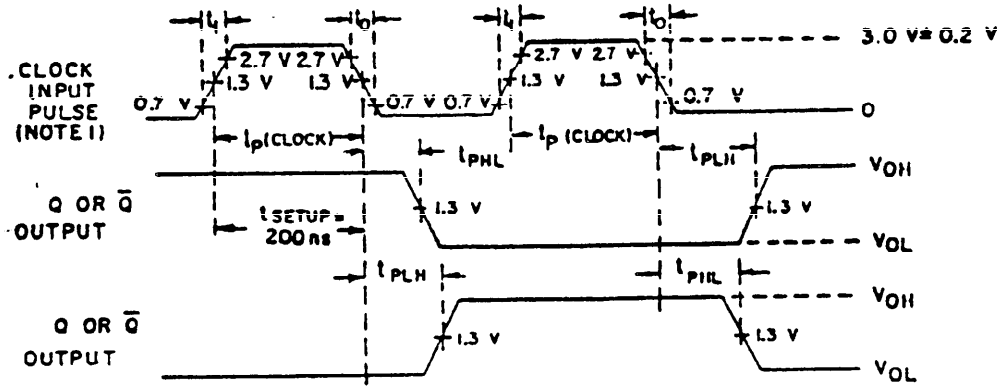
NOTES:

1. Clear input pulse characteristics: V_{gen} = 3.0 V ±0.2 V, t_0 = 15 ns, t_1 = 15 ns, t_p (clear) = 100 ns, PRR = 0.5 MHz and Z_{out} =50Ω.
2. C_L = 50 pF minimum and includes probe and jig capacitance.
3. R_L = 4 kΩ ±5% and C_1 =30 pF minimum.
4. All diodes are 1N916 or equivalent.
5. Clock input pulse characteristics: V_{gen} = 3.0 V ±0.2 V, t_p (clock) ≥ 200 ns, PRR = 0.5 MHz.

FIGURE 7. Clear switching test circuit for device type 03.

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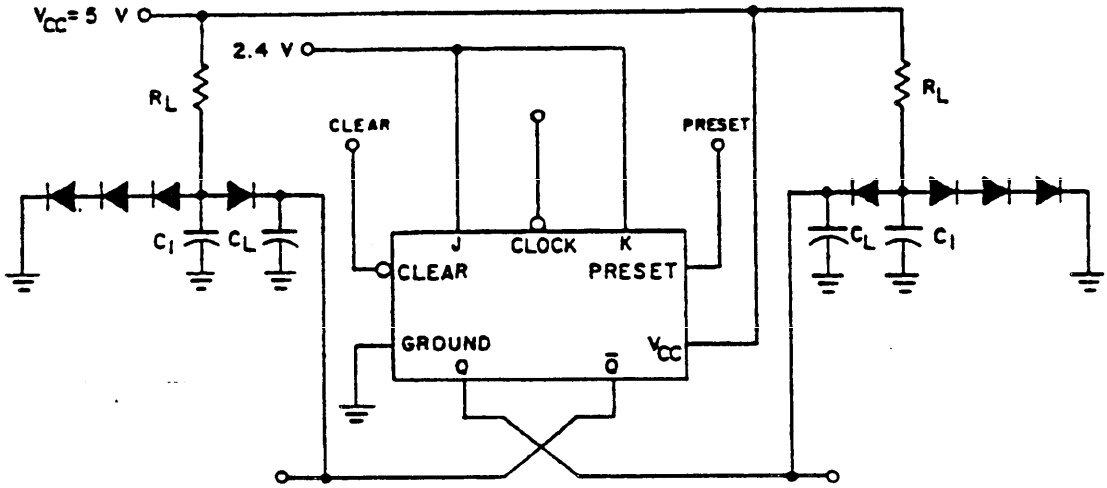


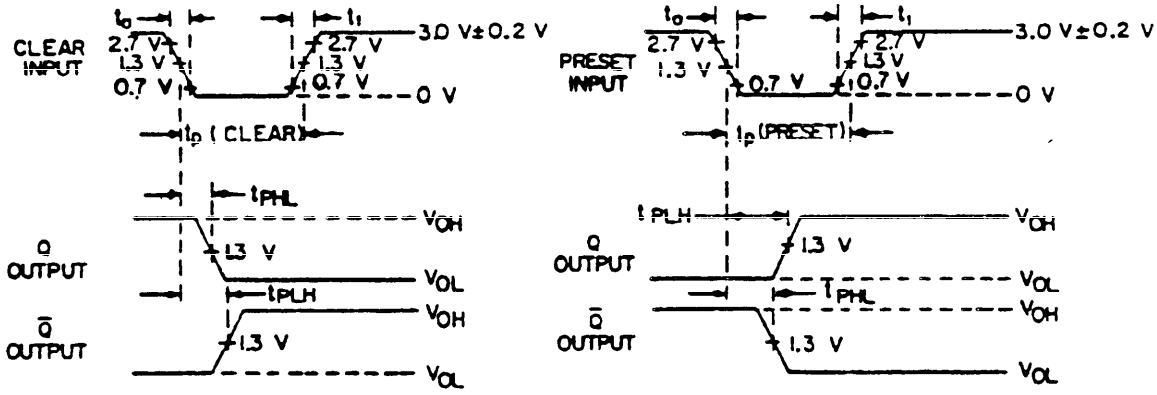


NOTES:
1. Clock input pulse characteristics: V_{gen} = 3.0 V ± 0.2 V, t_0 = 15 ns, tp = 200 ns, and PRR = 0.5 MHz, when testing F_{max}, PRR = see table III.
2. All diodes are 1N916 or equivalent.
3. C_L = 50 pF minimum and includes probe and jig capacitance.
4. R_L = 4 kΩ ±5% and C_l = 30 pF minimum.

FIGURE 8 Synchronous switching test circuit for device type 03.

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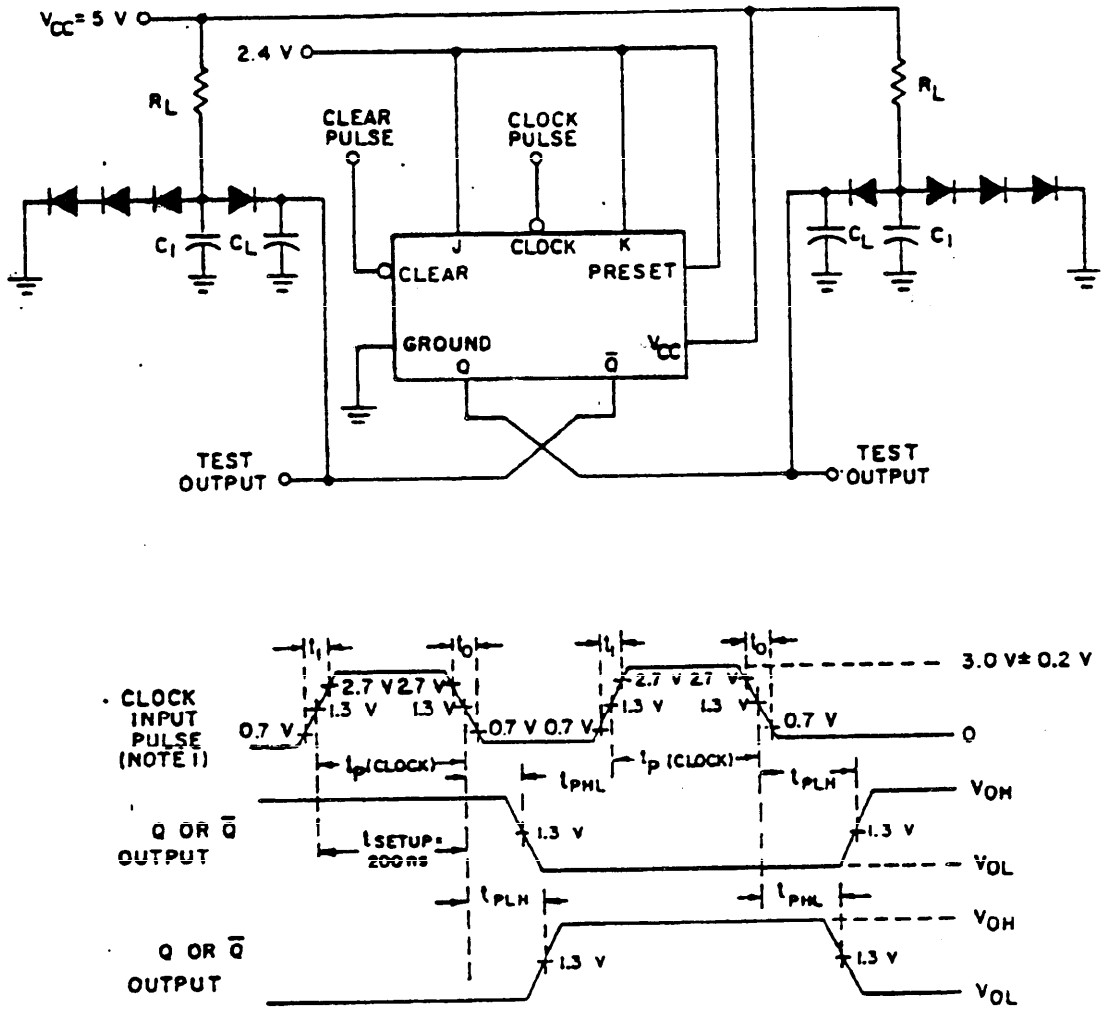


NOTES:

1. Clear or preset input pulse characteristics: V_{gen} = 3.0 V ± 0.2 V, t_0 = 15 ns, t_1 = 15 ns, t_p (Clear) = t_p (clear) = t_p (preset) = 100 ns, PRR = 0.5 MHz and $Z_{out} \approx 50\Omega$
2. C_L = 50 pF minimum and includes probe and jig capacitance.
3. R_L = 4 kΩ ± 5% and C_l = 30 pF minimum.
4. All diodes are 1N916 or equivalent.
5. When testing clear to output switching, preset input shall have a negative pulse; when testing preset to output switching, clear input shall have a negative pulse (see table III).

FIGURE 9. Clear and preset switching test circuit for device type 04.

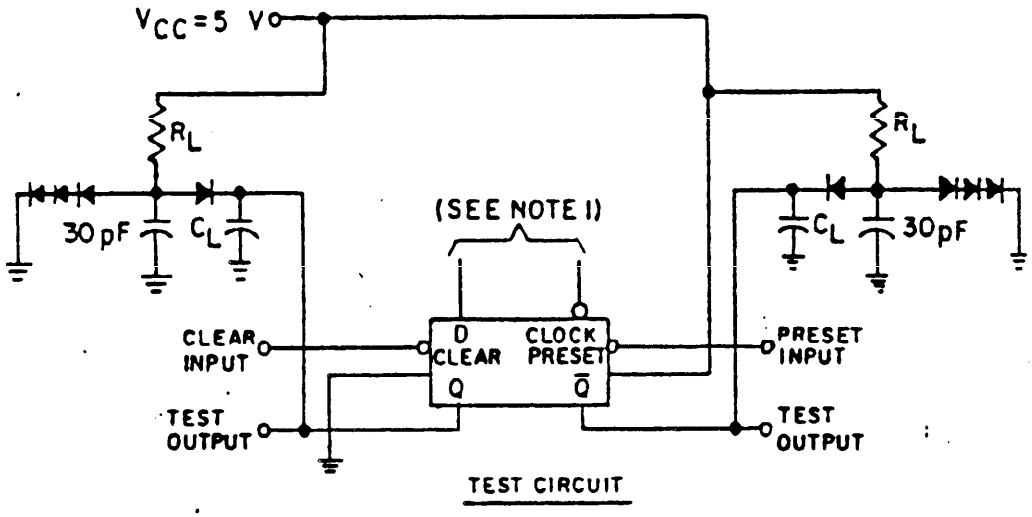
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Check the source to verify that this is the current version before use.



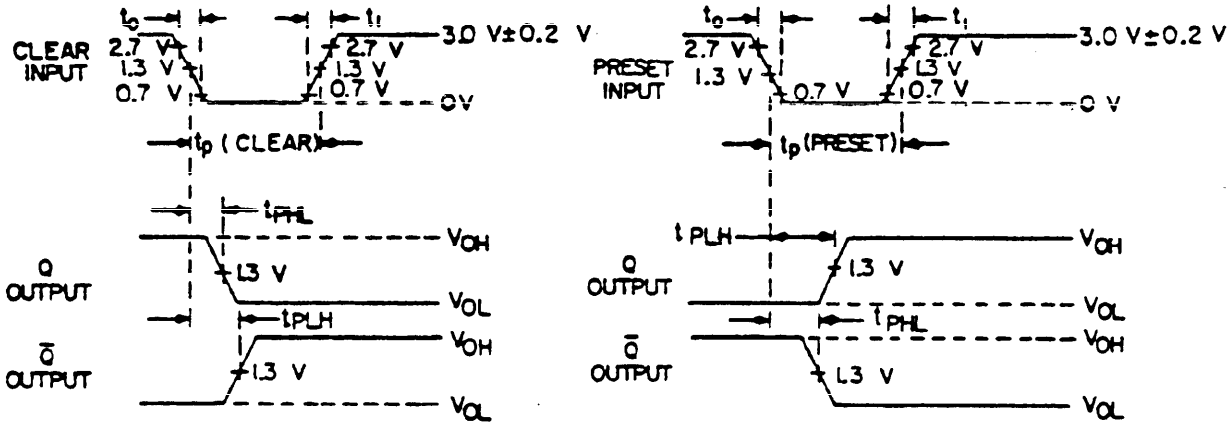
NOTES:
1. Clock input pulse characteristics: V_{gen} = 3.0 V ±0.2V. t_0 = 15 ns, t_p = 200 ns, and PRR = 0.5 MHz, when testing F_{max}, PRR = see table III.
2. All diodes are 1N916 or equivalent.
3. C_L = 50 pF minimum and includes probe and jig capacitance.
4. R_L = 4 kΩ ±5% and C_l = 30 pF minimum.

FIGURE 10. Synchronous switching test circuit for device type 04.

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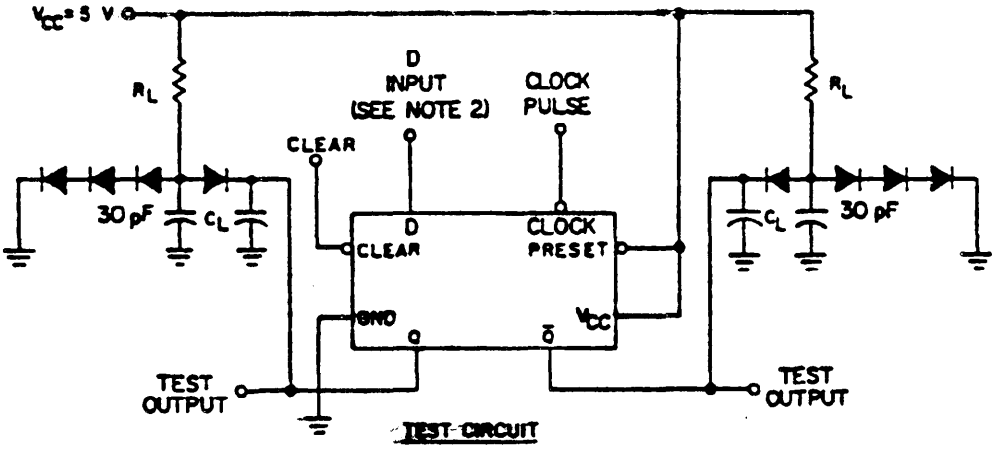
TEST CIRCUIT



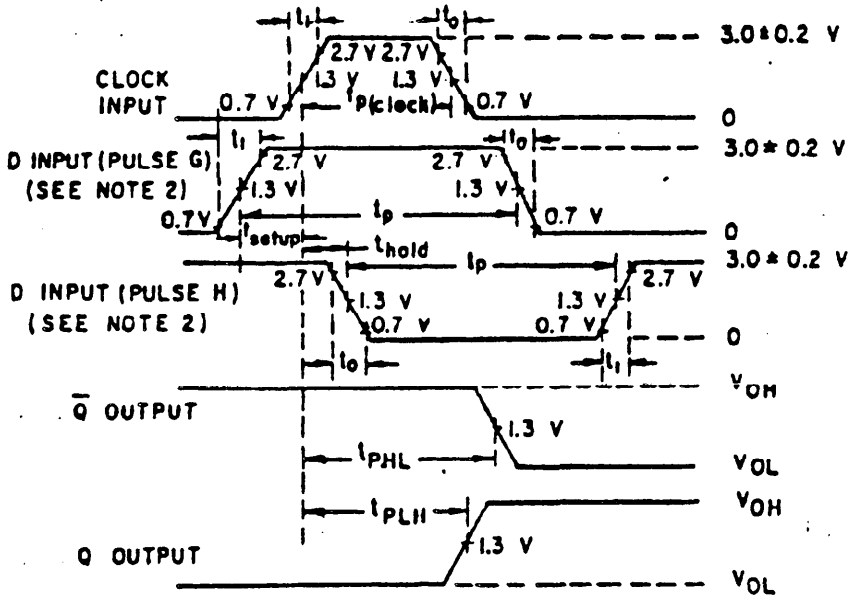
NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or D inputs.
2. All diodes are 1N916 or equivalent.
3. Clear or preset input pulse characteristics: V_{gen} = 3.0 ± 0.2 V, t_0 = 15 ns, t_p = 100 ns, PRR = 0.5 MHz.
4. C_L = 50 pF minimum and includes probe and jig capacitance.
5. R_L = 4 kΩ ±5%.
6. When testing clear to output switching, preset input shall have a negative pulse; when testing preset to output switching, clear input shall have a negative pulse (see Table III).

FIGURE 11. Clear and preset switching test circuit and waveforms for device type 05.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:

1. Clock input pulse has the following characteristics:

   V_{gen} = 3.0 ± 0.2 V, t_0 = 15 ns, tl = 15 ns, t_p = 200 ns,

   and PRR = 0.5 MHz.

   When testing F_{max}, PRR = see table III.

2. D input (pulse G) has the following characteristics:

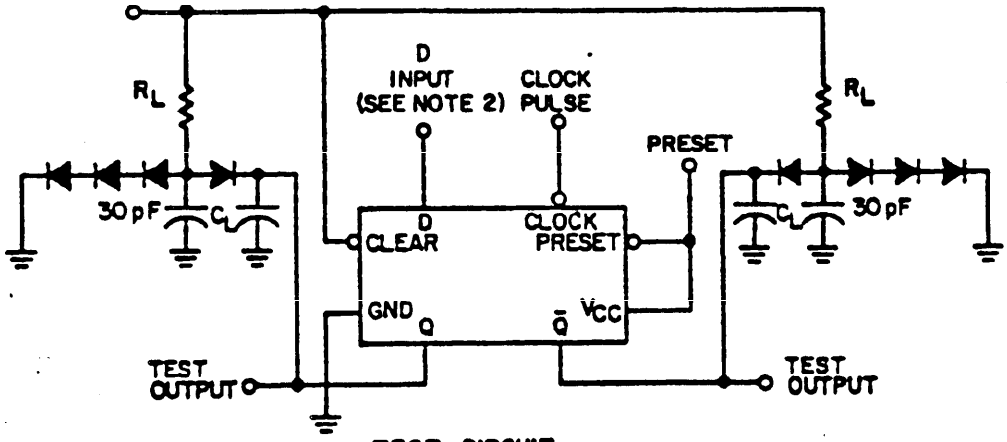
   V_{gen} 3.0 ± 0.2 V, t_0 = 15 ns, t_{setup} = 50 ns, t_1 = 15 ns,

   t_p = 100 ns, and PRR is 50% of the clock PRR.

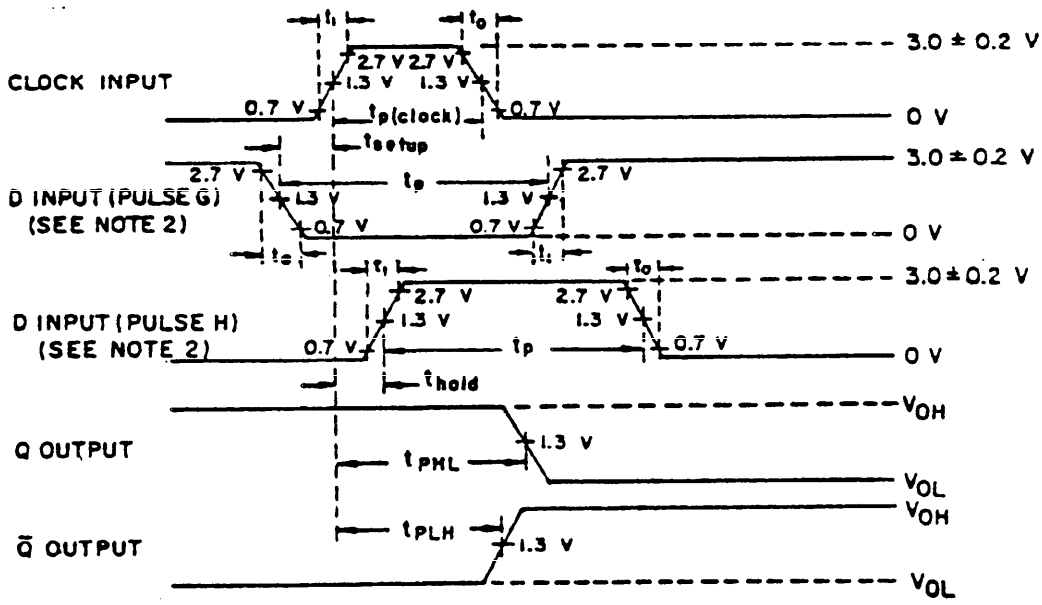
3. All diodes are 1N916 or equivalent.
4. C_L = 50 pf Minimum and includes probe and jig capacitance.
5. R_L = 4kΩ ±5%.

FIGURE 12. Synchronous switching test circuit for device type 05.

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TEST CIRCUIT



VOLTAGE WAVEFORM

NOTES:

1. Clock input pulse has the following characteristics: V_{gen} = 3.0 ±0.2 V, t_0 = 15 ns, t_p = 200 ns, and PRR = 0.5 MHz. When testing F_{max}, PRR see table III.
2. D input (pulse G) has the following characteristics: V_{gen} = 3.0 ±0.2 V, t_0 = 15 ns, t_{setup} = 50 ns, t_1 = 15 ns, t_p = 100 ns, and PRR is 50% of the clock PRR. D input (pulse H) has the following characteristics: V_{gen} 3.0 ±0.2 V, t_0 = 15 ns, t_{hold} = 0 ns, t_1 = 15 ns, t_p = 80 ns and PRR is 50% of the clock PRR.
3. All diodes are 1N916 or equivalent.
4. C_L = 50 pF minimum and includes probe and jig capacitance.
5. R_L = 4 kΩ ±5%.

FIGURE 13. Synchronous switching test circuit for device type 05.

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Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C Test no.	1 9 R_1	2 12 Clock	3 13 Preset	4 14 V_{CC}	5 2 Clear	6 1 NC	7 3 S_1	8 4 S_2	9 5 S_3	10 6 $\overline{Q}$	11 7 GND	12 8 Q	13 10 R_2	14 11 R_3	Measured terminal	Test limits Min	Test limits Max	Unit
T_C = +25°C	V_{OH}	3006	1	0.7 V	A	4.5 V	4.5 V	4.5 V		2.0 V	2.0 V	2.0 V		GND	-100 µA	0.7 V	0.7 V	Q	2.6		V
			2	2.0 V	A	4.5 V	"	4.5 V		0.7 V	0.7 V	0.7 V	-100 µA	"	"	2.0 V	2.0 V	$\overline{Q}$	"		"
			3	4.5 V	4.5 V	0.7 V	"	2.0 V		4.5 V	4.5 V	4.5 V		"	-100 µA	4.5 V	4.5 V	Q	"		"
			4	4.5 V	4.5 V	2.0 V	"	0.7 V		4.5 V	4.5 V	4.5 V	-100 µA	"		4.5 V	4.5 V	$\overline{Q}$	"		"
	V_{OL}	3007	5	2.0 V	A	4.5 V	"	4.5 V		0.7 V	0.7 V	0.7 V		"	2 mA	2.0 V	2.0 V	Q		0.3	"
			6	0.7 V	A	4.5 V	"	4.5 V		2.0 V	2.0 V	2.0 V	2 mA	"		0.7 V	0.7 V	$\overline{Q}$		"	"
			7	4.5 V	4.5 V	0.7 V	"	2.0 V		4.5 V	4.5 V	4.5 V	2 mA	"		4.5 V	4.5 V	Q		"	"
			8	4.5 V	"	2.0 V	"	0.7 V		4.5 V	"	"		"	2 mA	4.5 V	4.5 V	$\overline{Q}$		"	"
	I_{IL1}	3009	9		"		5.5 V	8		0.3 V	"	"		"				S_1	-43	-140	µA
			10		"		"	"		4.5 V	0.3 V	"		"				S_2	"	"	"
			11		"		"			4.5 V	4.5 V	0.3 V		"				S_3	"	"	"
			12	0.3 V	"	8	"							"		4.5 V	4.5 V	R_1	"	"	"
			13	4.5 V	"	"	"							"		0.3 V	4.5 V	R_2	"	"	"
			14	"	"		"							"		4.5 V	0.3 V	R_3	"	"	"
	I_{IL2}		15	"	0.3 V	"	"			4.5 V	4.5 V	4.5 V		"		"	4.5 V	Clock	-120	-360	"
			16	"	0.3 V		"	8		"	"	"		"		"	"	Clock	-120	-360	"
			17	"		0.3 V	"			"	"	"		"		"	"	Preset	-86	-280	"
			18	"			"	0.3 V		"	"	"		"		"	"	Clear	-86	-280	"
	I_{IH1}	3010	19		GND		"	GND		2.4 V	GND	GND		"				S_1		10	"
			20		"		"	"		GND	2.4 V	GND		"				S_2		"	"
			21		"		"			GND	GND	2.4 V		"				S_3		"	"
			22	2.4 V	"	GND	"							"		GND	GND	R_1		"	"
			23	GND	"	"	"							"		2.4 V	GND	R_2		"	"
			24	GND	"	"	"							"		GND	2.4 V	R_3		"	"
	I_{IH2}		25		"		"	GND		5.5 V	GND	GND		"				S_1		100	"
			26		"		"	"		GND	5.5 V	GND		"				S_2		"	"
			27		"		"			GND	GND	5.5 V		"				S_3		"	"
			28	5.5 V	"	GND	"							"		GND	GND	R_1		"	"
			29	GND	"	"	"							"		5.5 V	GND	R_2		"	"
			30	GND	"	"	"							"		GND	5.5 V	R_3		"	"
	I_{IH3}		31		"		"	2.4 V		GND	GND	GND	GND	"				Clear		20	"
	I_{IH3}		32	GND	"	2.4 V	"							"	GND	GND	GND	Preset		20	"
	I_{IH4}		33	GND	"	5.5 V	"							"				Preset		200	"
			34		"		"	5.5 V		GND	GND	GND	GND	"	GND	GND	GND	Clear		"	"
			35	GND	5.5 V	GND	"			"	"	"		"		GND	GND	Clock		"	"
			36	"	5.5 V		"	GND		"	"	"		"		"	"	Clock		"	"
	I_{IH10}		37	"	2.4 V	GND	"			"	"	"		"		"	"	Clock	0	-200	"
	I_{IH10}		38	"	2.4 V		"	GND		"	"	"		"		"	"	Clock	0	-200	"
	I_{OS}	3011	39	4.5 V	GND	GND	"			4.5 V	4.5 V	4.5 V		"	GND	4.5 V	4.5 V	Q	-3	-15	mA
	I_{OS}	3011	40	4.5 V	"		"	GND		4.5 V	4.5 V	4.5 V	GND	"		4.5 V	4.5 V	$\overline{Q}$	-3	-15	"
	I_{CC}	3005	41 CKT A	GND	"	GND	"	4.5 V		GND	GND	GND		"		GND	GND	V_{CC}		1.90	"
			41 CKT B	"	"	GND	"	4.5 V		"	"	"		"		"	"	"		1.44	"
			42 CKT A	"	"	4.5 V	"	GND		"	"	"		"		"	"	"		1.90	"
			42 CKT B	"	"	4.5 V	"	GND		"	"	"		"		"	"	"		1.44	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C.																				

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device 01 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Test no.																Min	Max	
7 T_C = +25°C 2/ 4/			43	A	A	A	4.5 V	0		A	A	A	H 3/		L 3/	A	A	All outputs			H or L as shown 3/
			44	B	A	A		A		B	A	A	H		L	A	A				
			45	B	B	A				B	A	A	L		H	A	A				
			46	A	B	A				A	A	A	H		L	B	B				
			47	A	B	A				A	B	A	L		H	B	A				
			48	A	A	B				A	B	A	H		L	B	A				
			49	A	A	B				A	B	A	L		H	B	A				
			50	B	A	B				B	A	A	H		L	A	B				
			51	B	A	B				B	A	A	L		H	A	B				
			52	B	B	B				B	B	A	H		L	A	A				
			53	B	B	B				B	B	A	L		H	A	A				
			54	A	B	B				A	B	A	H		L	B	A				
			55	A	B	B				A	B	A	L		H	B	A				
			56	B	A	B				B	A	A	H		L	A	B				
			57	B	A	B				B	A	A	L		H	A	B				
			58	A	B	B				A	A	A	H		L	B	B				
			59	A	B	B				A	A	A	L		H	B	B				
8 2/ 4/	Subgroup tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C																				
9 T_C = +25°C	f_{MAX} 5/ f_{MAX} 5/	(Fig. 6)	60 61	D	IN IN	5.0 V 5.0 V	5.0 V	B B		C	2.0 V	2.4 V	OUT	GND	OUT	2.0 V	2.4 V	Q $\overline{Q}$	3		MHz MHz
	t_{PLH}	3003 (Fig. 5)	62 CKT A		2.4 V	J		IN					OUT		OUT			Clear/$\overline{Q}$	10		ns
	"		62 CKT B			IN		IN					OUT					Clear/Q		75	
	"		63 CKT A			J		J										Preset/Q		150	
	"		63 CKT B			IN		IN										Preset/$\overline{Q}$		75	
	t_{PHL}		64 CKT A		GND	J		J										Clear/Q		150	
	"		64 CKT B			IN		IN										Clear/$\overline{Q}$		200	
	"		65 CKT A			J		J	5.0 V									Preset/$\overline{Q}$		90	
	"		65 CKT B		IN	J		J	5.0 V									Clear/$\overline{Q}$		200	
	t_{PLH}	3003 (Fig. 6)	66 CKT A			J		J	5.0 V				OUT		OUT			Preset/$\overline{Q}$		90	
	"		66 CKT B			IN		IN	5.0 V				OUT					Clock/$\overline{Q}$		75	
	"		67 CKT A			J		J	5.0 V									Clock/Q		50	
	"		67 CKT B			IN		IN	5.0 V									Clock/$\overline{Q}$		75	
	t_{PHL}		68 CKT A			J		J	5.0 V									Clock/Q		50	
	"		68 CKT B			IN		IN	5.0 V									Clock/$\overline{Q}$		150	
	"		69 CKT A			J		J	5.0 V				OUT		OUT			Clock/Q		150	
	"		69 CKT B			IN		IN	5.0 V				OUT					Clock/$\overline{Q}$		70	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 02 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D: Case C: Test no.	1: 9: R_1	2: 12: Clock	3: 13: Preset	4: 14: V_{CC}	5: 2: Clear	6: 1: NC	7: 3: S_1	8: 4: S_2	9: 5: S_3	10: 6: $\overline{Q}$	11: 7: GND	12: 8: Q	13: 10: R_2	14: 11: R_3	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	f_{MAX} 5/ f_{MAX} 5/	(Fig. 6)	70 71	D "	IN IN	5.0 V 5.0 V	5.0 V "	B B		C "	2.4 V "	2.4 V "	OUT	GND "	OUT	2.4 V "	2.4 V "	Q $\overline{Q}$	2.5 2.5		MHz MHz
	t_{PLH}	3003 (Fig. 5)	*72 CKT A	"	2.4 V	J	"	IN		"	"	"	OUT	"		"	"	Clear/$\overline{Q}$	10	125	ns
	"		*72 CKT B	"	"	J	"	IN		"	"	"	OUT	"		"	"	Clear/$\overline{Q}$	"	65	"
	"		*73 CKT A	"	"	IN	"	J		"	"	"		"	OUT	"	"	Preset/Q	"	125	"
	"		*73 CKT B	"	"	IN	"	J		"	"	"		"	"	"	"	Preset/Q	"	65	"
	t_{PHL}		74 CKT A	"	GND	J	"	IN		"	"	"		"	"	"	"	Clear/Q	"	250	"
	"		74 CKT B	"	"	J	"	IN		"	"	"		"	"	"	"	Clear/Q	"	100	"
	"		75 CKT A	"	"	IN	"	J		"	"	"	OUT	"		"	"	Preset/$\overline{Q}$	"	250	"
	"		75 CKT B	"	"	IN	"	J		"	"	"	"	"		"	"	Preset/$\overline{Q}$	"	100	"
	t_{PLH}	3003 (Fig. 6)	76 CKT A	"	IN	J	"	5.0 V		"	"	"	"	"		"	"	Clock/$\overline{Q}$	"	125	"
	"		76 CKT B	"	"	J	"	5.0 V		"	"	"	"	"	OUT	"	"	Clock/$\overline{Q}$	"	65	"
	"		77 CKT A	"	"	5.0 V	"	J		"	"	"		"	"	"	"	Clock/Q	"	125	"
	"		77 CKT B	"	"	5.0 V	"	J		"	"	"		"	"	"	"	Clock/Q	"	65	"
	t_{PHL}		78 CKT A	"	"	J	"	5.0 V		"	"	"		"	"	"	"	Clock/Q	"	200	"
	"		78 CKT B	"	"	J	"	5.0 V		"	"	"		"		"	"	Clock/Q	"	85	"
	"		79 CKT A	"	"	5.0 V	"	J		"	"	"	OUT	"		"	"	Clock/$\overline{Q}$	"	200	"
	"		79 CKT B	"	"	5.0 V	"	J		"	"	"	OUT	"		"	"	Clock/$\overline{Q}$	"	85	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

NOTE: A = normal clock pulse, B = momentary GND, then 4.5 V,
C = input connected to $\overline{Q}$, D = input connected to Q,
J = input pulse $t_p \geq 100$ ns, PRR = 0.5 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Tests shall be performed in sequence.
3/ Input voltages shown are: A = 2.4 V minimum and B = 0.4 V maximum.
4/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L $\leq$ 1.5 V when using a high speed checker single comparator.
5/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
* These tests are performed at device manufacturer's option.

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Subgroup	Symbol	MIL-STD-883 method	Cases K, L, 9 Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Test no.	9	12	13	14	2	1	3	4	5	6	7	8	10	11		Min	Max	
				K_1	Clock	Preset	V_{CC}	Clear	NC	J_1	J_2	J_3	$\overline{Q}$	GND	Q	K_2	K_3				
T_C = 1 +25°C	V_{OH}	3006	1	0.7 V	A	4.5 V	4.5 V	4.5 V		2.0 V	2.0 V	2.0 V		GND	-100 µA	0.7 V	0.7 V	Q	2.4		V
			2	2.0 V	A	4.5 V	"	4.5 V		0.7 V	0.7 V	0.7 V	-100 µA	"	"	2.0 V	2.0 V	$\overline{Q}$	"		"
			3	4.5 V	4.5 V	0.7 V	"	2.0 V		4.5 V	4.5 V	4.5 V	"	"	-100 µA	4.5 V	4.5 V	Q	"		"
			4	4.5 V	"	2.0 V	"	0.7 V		4.5 V	4.5 V	4.5 V	-100 µA	"	"	4.5 V	4.5 V	$\overline{Q}$	"		"
	V_{OL}	3007	5	2.0 V	A	4.5 V	"	4.5 V		0.7 V	0.7 V	0.7 V		"	2 mA	2.0 V	2.0 V	Q		0.3	"
			6	0.7 V	A	4.5 V	"	4.5 V		2.0 V	2.0 V	2.0 V	2 mA	"	"	0.7 V	0.7 V	$\overline{Q}$		"	"
			7	4.5 V	4.5 V	0.7 V	"	2.0 V		4.5 V	4.5 V		2 mA	"	"	4.5 V	4.5 V	$\overline{Q}$		"	"
			8	4.5 V	"	2.0 V	"	0.7 V		4.5 V	"		"	"	2 mA	4.5 V	4.5 V	Q		"	"
	I_{IL1}	3009	9				5.5 V	8		0.3 V	"	4.5 V		"				J_1	-43	-140	µA
			10				"	"		4.5 V	0.3 V	4.5 V		"				J_2	"	"	"
			11				"			4.5 V	4.5 V	0.3 V		"				J_3	"	"	"
			12	0.3 V	"	8	"							"		4.5 V	4.5 V	K_1	"	"	"
			13	4.5 V	"	"	"							"		0.3 V	4.5 V	K_2	"	"	"
			14		"		"							"		4.5 V	0.3 V	K_3	"	"	"
	I_{IL2}		15	"	0.3 V	"	"	8		4.5 V	4.5 V	4.5 V		"		"	4.5 V	Clock	-105	-360	"
			16	"	0.3 V	0.3 V	"			"	"	"		"		"	"	Clock	-105	-360	"
			17	"			"	0.3 V		"	"	"		"		"	"	Preset	-86	-280	"
			18	"			"			"	"	"		"		"	"	Clear	-86	-280	"
	I_{IH1}	3010	19		GND			GND		2.4 V	GND	GND						J_1		10	
			20		"			"		GND	2.4 V	GND						J_2		"	
			21		"					GND	GND	2.4 V						J_3		"	
			22	2.4 V	"	GND										GND	GND	K_1		"	
			23	GND	"	"										2.4 V	GND	K_2		"	
			24	GND	"	"										GND	2.4 V	K_3		"	
	I_{IH2}		25					GND		5.5 V	GND	GND						J_1		100	
			26					"		GND	5.5 V	GND						J_2		"	
			27							GND	GND	5.5 V						J_3		"	
			28	5.5 V	"	GND										GND	GND	K_1		"	
			29	GND	"	"										5.5 V	GND	K_2		"	
			30	GND	"	"										GND	5.5 V	K_3		"	
	I_{IH3}		31					2.4 V		GND	GND	GND	GND					Clear		20	
	I_{IH3}		32	GND	"	2.4 V									GND	GND	GND	Preset		20	
	I_{IH4}		33	GND		5.5 V		5.5 V		GND	GND	GND	GND		GND	GND	GND	Preset		200	
			34													GND		Clear			
			35	GND	5.5 V	GND		GND									GND	Clock			
			36	"	5.5 V													Clock			
	I_{IH10}		37		2.4 V	GND												Clock	0	-200	
	I_{IH10}		38		2.4 V			GND										Clock	0	-200	
	I_{OS}	3011	39	4.5 V	GND	GND				4.5 V	4.5 V	4.5 V	GND		GND	4.5 V	4.5 V	Q	-3	-15	mA
	I_{OS}	3011	40	4.5 V	"			GND		4.5 V	4.5 V	4.5 V				4.5 V	4.5 V	$\overline{Q}$	-3	-15	"
	I_{CC}	3005	41 CKT A	GND		GND		4.5 V		GND	GND	GND				GND	GND	V_{CC}		1.90	
			41 CKT B			GND		4.5 V												1.44	
			42 CKT A			4.5 V		GND												1.90	
			42 CKT B			4.5 V		GND												1.44	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C.																				

See footnotes at end of device type 02.

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Table III. Group A inspection for device ... continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Case C	9	12	13	16	2	1	3	4	5	6	7	8	10	11		Min	Max	
			Test no.	K_1	Clock	Preset	V_{CC}	Clear	NC	J_1	J_2	J_3	$\overline{Q}$	GND	Q	K_2	K_3				
7 T_C = +25°C 2/ 6/			43	B	B	A	4.5 V	B	B	B	B	B	H 3/	GND	L 3/	B	B	All outputs	H or L as shown 3/		
			44	-	B	-	-	A	-	-	-	-	L	-	H	B	B				
			45	-	A	-	-	-	-	-	-	-	-	-	-	A	A				
			46	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
			47	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			48	A	-	-	-	-	-	-	-	-	-	-	-	B	-				
			49	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			50	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			51	-	B	-	-	-	-	-	-	-	-	-	-	A	B				
			52	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			53	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			54	-	-	-	-	B	-	-	-	-	H	-	L	-	-				
			55	B	-	-	-	A	-	-	A	A	-	-	-	B	-				
			56	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			57	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			58	-	B	-	-	-	-	A	B	-	-	-	-	-	-				
			59	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			60	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			61	-	B	-	-	-	-	-	A	B	-	-	-	-	-				
			62	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			63	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			64	A	B	-	-	-	-	-	-	A	-	-	-	A	A				
			65	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			66	-	B	-	-	-	-	-	-	-	L	-	H	-	-				
			67	-	A	-	-	-	-	-	-	-	L	-	H	-	-				
			68	-	B	-	-	-	-	-	-	-	H	-	L	-	-				
			69	-	B	-	-	B	-	-	-	-	-	-	-	-	-				
			70	-	A	-	-	-	-	-	-	-	-	-	-	-	-				
			71	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			72	-	B	B	-	-	-	-	-	-	-	-	H	-	-				
			73	A	A	-	-	-	-	-	-	-	-	-	-	-	-				
			74	-	B	-	-	-	-	-	-	-	-	-	-	-	-				
			75	-	A	-	-	A	-	-	-	-	L	-	-	-	-				
			76	-	-	A	-	-	-	-	-	-	-	-	-	-	-				
			77	B	-	-	-	-	-	-	-	-	-	-	-	-	-				
			78	-	-	-	-	-	-	B	-	-	-	-	-	-	-				
			79	-	B	-	-	-	-	B	A	-	H	-	L	-	-				
			80	A	A	-	-	-	-	A	B	-	-	-	-	-	-				
			81	A	A	-	-	-	-	B	-	-	-	-	-	-	-				
8 2/3/4/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C																				

See footnotes at end of device type 02.

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Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C Test no.	1 K_1	2 Clock	3 Preset	4 V_{CC}	5 Clear	6 NC	7 J_1	8 J_2	9 J_3	10 $\overline{Q}$	11 GND	12 Q	13 K_2	14 K_3	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	f_{MAX} 5/	(Fig. 6)	82	2.4 V	IN	B	5.0 V	5.0 V		2.4 V	2.4 V	2.4 V		GND	OUT	2.4 V	2.4 V	Q	3		MHz
	f_{MAX} 5/		83	"	IN	B	"	5.0 V		"	"	"	OUT	"	"	"	"	$\overline{Q}$	3		MHz
	t_{PLH}	3003 (Fig. 5)	*84 CKT A	"	2.4 V	J	"	IN		"	"	"	OUT	"		"	"	Clear/$\overline{Q}$	10	75	ns
	"		*84 CKT B	"	"	J	"	IN		"	"	"	OUT	"		"	"	Clear/$\overline{Q}$	"	50	"
	"		*85 CKT A	"	"	IN	"	J		"	"	"	"	"	OUT	"	"	Preset/Q	"	75	"
	"		*85 CKT B	"	"	IN	"	J		"	"	"	"	"	"	"	"	Preset/Q	"	50	"
	t_{PHL}		86 CKT A	"	GND	J	"	IN		"	"	"	"	"	"	"	"	Clear/Q	"	200	"
	"		86 CKT B	"	"	J	"	IN		"	"	"	"	"	"	"	"	Clear/Q	"	90	"
	"		87 CKT A	"	"	IN	"	J		"	"	"	OUT	"	"	"	"	Preset/$\overline{Q}$	"	200	"
	"		87 CKT B	"	"	IN	"	J		"	"	"	"	"	"	"	"	Preset/$\overline{Q}$	"	90	"
	t_{PLH}	3003 (Fig. 6)	88 CKT A	"	IN	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/$\overline{Q}$	"	75	"
	"		88 CKT B	"	"	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/$\overline{Q}$	"	50	"
	"		89 CKT A	"	"	5.0 V	"	J		"	"	"	"	"	OUT	"	"	Clock/Q	"	75	"
	"		89 CKT B	"	"	5.0 V	"	J		"	"	"	"	"	"	"	"	Clock/Q	"	50	"
	t_{PHL}		90 CKT A	"	"	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/Q	"	150	"
	"		90 CKT B	"	"	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/Q	"	70	"
	"		91 CKT A	"	"	5.0 V	"	J		"	"	"	OUT	"	"	"	"	Clock/$\overline{Q}$	"	150	"
	"		91 CKT B	"	"	5.0 V	"	J		"	"	"	OUT	"	"	"	"	Clock/$\overline{Q}$	"	70	"
10 T_C = +125°C	f_{MAX} 5/	(Fig. 6)	92	"	"	B	"	5.0 V		"	"	"		"	OUT	"	"	Q	2.5		MHz
	f_{MAX} 5/		93	"	"	B	"	5.0 V		"	"	"	OUT	"		"	"	$\overline{Q}$	2.5		MHz
	t_{PLH}	3003 (Fig. 5)	*94 CKT A	"	2.4 V	J	"	IN		"	"	"	OUT	"		"	"	Clear/$\overline{Q}$	10	125	ns
	"		*94 CKT B	"	"	J	"	IN		"	"	"	OUT	"		"	"	Clear/$\overline{Q}$	"	65	"
	"		*95 CKT A	"	"	IN	"	J		"	"	"	"	"	OUT	"	"	Preset/Q	"	125	"
	"		*95 CKT B	"	"	IN	"	J		"	"	"	"	"	"	"	"	Preset/Q	"	65	"
	t_{PHL}		96 CKT A	"	GND	J	"	IN		"	"	"	"	"	"	"	"	Clear/Q	"	250	"
	"		96 CKT B	"	"	J	"	IN		"	"	"	"	"	"	"	"	Clear/Q	"	100	"
	"		97 CKT A	"	"	IN	"	J		"	"	"	OUT	"	"	"	"	Preset/$\overline{Q}$	"	250	"
	"		97 CKT B	"	"	IN	"	J		"	"	"	"	"	"	"	"	Preset/$\overline{Q}$	"	100	"
	t_{PLH}	3003 (Fig. 6)	98 CKT A	"	IN	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/$\overline{Q}$	"	125	"
	"		98 CKT B	"	"	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/$\overline{Q}$	"	65	"
	"		99 CKT A	"	"	5.0 V	"	J		"	"	"	"	"	OUT	"	"	Clock/Q	"	125	"
	"		99 CKT B	"	"	5.0 V	"	J		"	"	"	"	"	"	"	"	Clock/Q	"	65	"
	t_{PHL}		100 CKT A	"	"	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/Q	"	200	"
	"		100 CKT B	"	"	J	"	5.0 V		"	"	"	"	"	"	"	"	Clock/Q	"	85	"
	"		101 CKT A	"	"	5.0 V	"	J		"	"	"	OUT	"	"	"	"	Clock/$\overline{Q}$	"	200	"
	"		101 CKT B	"	"	5.0 V	"	J		"	"	"	OUT	"	"	"	"	Clock/$\overline{Q}$	"	85	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

NOTE: A = normal clock pulse, B = momentary GND, then 4.5 V,
J = input pulse $t_p \geq 100$ ns, PRR = 0.5 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Tests shall be performed in sequence.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L $\leq$ 1.5 V when using a high speed checker single comparator.
4/ Input voltages shown are: A = 2.1 V minimum and B = 0.4 V maximum.
5/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
* These tests are performed at device manufacturer's option.

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TABLE III. Group A inspection for device type 03. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	Clock 1	Clear 1	K_1	V_{CC}	Clock 2	Clear 2	J_2	$\overline{Q}_2$	Q_2	K_2	GND	Q_1	$\overline{Q}_1$	J_1				
1 T_C = +25°C	V_{OH}	3006	1	A	4.5 V	0.7 V	4.5 V							GND	-100 µA	-100 µA	2.8 V	$\overline{Q}_1$	2.4		V
			2	A	4.5 V	2.0 V	"							"		-100 µA	0.7 V	Q_1	"		"
			3	4.5 V	0.7 V	4.5 V	"							"		-100 µA	4.5 V	J_1	"		"
			4				"	A	4.5 V	2.0 V		-100 µA	0.7 V	"				K_1	"		"
			5				"	A	4.5 V	0.7 V	-100 µA		2.0 V	"				J_2	"		"
			6				"	4.5 V	0.7 V	4.5 V	-100 µA		4.5 V	"				K_2	"		"
	V_{OL}	3007	7	A	4.5 V	2.0 V	"							"	2 mA	2 mA	0.7 V	$\overline{Q}_1$		0.3	"
			8	A	4.5 V	0.7 V	"							"			2.8 V	Q_1		"	"
			9	4.5 V	0.7 V	4.5 V	"							"	2 mA		4.5 V	J_1		"	"
			10				"	A	4.5 V	0.7 V		2 mA	2.0 V	"				K_1		"	"
			11				"	A	4.5 V	0.7 V	2 mA		0.7 V	"				J_2		"	"
			12				"	4.5 V	0.7 V	4.5 V		2 mA	4.5 V	"				K_2		"	"
	I_{IL1}	3009	13	4.5 V	4.5 V		5.5 V							"	E	E	0.3 V	J_1	-43	-140	µA
			14	4.5 V	4.5 V	0.3 V	"							"				K_1	"	"	"
			15				"	4.5 V	4.5 V	0.3 V		E		"				J_2	"	"	"
			16				"	4.5 V	4.5 V		E		0.3 V	"				K_2	"	"	"
	I_{IL2}		17	4.5 V	0.3 V		"							"			4.5 V	Clear 1	-86	-280	"
			18	0.3 V	B	4.5 V	"							"			4.5 V	Clock 1	-120	-360	"
			19				"	4.5 V	0.3 V	4.5 V				"				Clear 2	-86	-280	"
			20				"	0.3 V	B	4.5 V			4.5 V	"				Clock 2	-120	-360	"
	I_{IH1}	3010	21	GND	GND		"							"			2.4 V	J_1		10	"
			22	GND	B	2.4 V	"							"				K_1		"	"
			23				"	GND	GND	2.4 V				"				J_2		"	"
			24				"	GND	B				2.4 V	"				K_2		"	"
	I_{IH2}		25	GND	GND		"							"			5.5 V	J_1		100	"
			26	GND	B	5.5 V	"							"				K_1		"	"
			27				"	GND	GND	5.5 V				"				J_2		"	"
			28				"	GND	B				5.5 V	"				K_2		"	"
	I_{IH3}		29	GND	2.4 V		"							"		GND	GND	Clear 1		20	"
	I_{IH3}		30				"	GND	2.4 V	GND	GND			"				Clear 2		20	"
	I_{IH4}		31	5.5 V	GND	GND	"							"			GND	Clock 1		200	"
			32	GND	5.5 V		"							"		GND	GND	Clear 1		"	"
			33				"	5.5 V	GND	GND			GND	"				Clock 2		"	"
			34				"	GND	5.5 V	GND	GND			"				Clear 2		"	"
	I_{IH10}		35	2.4 V	GND	GND	"							"			GND	Clock 1	0	-200	"
	I_{IH10}		36				"	2.4 V	GND	GND			GND	"				Clock 2	0	-200	"
	I_{OS}	3011	37	2.4 V	GND	2.4 V	"							"		GND	2.4 V	$\overline{Q}_1$	-3	-15	mA
		3011**	38	A	2.4 V	GND	"							"	GND		2.4 V	Q_1	"	"	"
		3011**	39				"	A	2.4 V	2.4 V		GND	GND	"				$\overline{Q}_2$	"	"	"
		3011	40				"	2.4 V	GND	2.4 V	GND		2.4 V	"				Q_2	"	"	"
	I_{CC}	3005	41 CKT A	F	4.5 V	GND	"	F	4.5 V	4.5 V			GND	"			4.5 V	V_{CC}		3.8	"
			41 CKT B	F	4.5 V	"	"	F	4.5 V	4.5 V			"	"			4.5 V	"		2.88	"
			42 CKT A	GND	GND	"	"	GND	GND	GND			"	"			GND	"		3.8	"
			42 CKT B	GND	GND	"	"	GND	GND	GND			"	"			GND	"		2.88	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C.																				

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device 03 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C / Test no.	1 / 1 / Clock 1	2 / 2 / Clear 1	3 / 3 / K_1	4 / 4 / V_{CC}	5 / 5 / Clock 2	6 / 6 / Clear 2	7 / 7 / J_2	8 / 8 / $\overline{Q}_2$	9 / 9 / Q_2	10 / 10 / K_2	11 / 11 / GND	12 / 12 / Q_1	13 / 13 / $\overline{Q}_1$	14 / 14 / J_1	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C 2/ 4/			43	B	B	B	4.5 V	B	B	A	H 3/	L 3/	B	GND	L 3/	H 3/	A	All outputs	H or L as shown 3/		
			44	A				A													
			45	B				B													
			46	B		A		B					A								
			47	A				A													
			48	B				B													
			49	B	A			B	A												
			50	A				A													
			51	B				B			L	H			H	L					
			52	A				A			L	H			H	L					
			53	B				B			H	L			L	H					
			54	A				A			H	L			L	H					
			55	B				B			L	H			H	L					
			56	B		B		B		B			B				B				
			57	A				A													
			58	B				B													
			59		B				B		H	L			L	H					
			60		A				A												
			61																		
			62		B																
			63							A							A				
			64		A																
			65		B						L	H			H	L					
			66		B	A				B			A				B				
			67		A																
			68		B					A	H	L			L	H					
			69		A												A				
			70							B			B				B				
			71		A																
			72			B				A	L	H			H	L					
			73		B								B				A				
			74		A					B			A				B				
			75			A															
			76																		
			77		B					A	H	L			L	H					
			78		A								B				A				
			79		B					B			A				B				
			80		A																
			81								L	H			H	L					
8 2/ 4/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C																				

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device 03 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	Clock 1	Clear 1	K_1	V_{CC}	Clock 2	Clear 2	J_2	$\overline{Q}_2$	Q_2	K_2	GND	Q_1	$\overline{Q}_1$	J_1				
9 T_C = +25°C	f_{MAX} 5/	(Fig. 8)	82	IN	B	2.4 V	5.0 V							GND	OUT		2.4 V	Clock/Q_1	3		MHz
			83	IN	B	2.4 V	"							"		OUT	2.4 V	Clock/$\overline{Q}_1$	"		"
			84				"	IN	B	2.4 V		OUT	2.4 V	"				Clock/Q_2	"		"
			85				"	IN	B	2.4 V	OUT		2.4 V	"				Clock/$\overline{Q}_2$	"		"
	t_{PLH}	3003 (Fig. 7)	*86 CKT A	IN	IN	GND	"							"		OUT	2.4 V	Clear/$\overline{Q}_1$	10	75	ns
			*86 CKT B	IN	IN	GND	"							"		OUT	2.4 V	Clear/$\overline{Q}_1$	"	50	"
			*87 CKT A				"	IN	IN	2.4 V	OUT		GND	"			"	Clear/$\overline{Q}_2$	"	75	"
			*87 CKT B				"	IN	IN	2.4 V	OUT		GND	"			"	Clear $\overline{Q}_2$	"	50	"
	t_{PHL}		88 CKT A	IN	IN	GND	"							"	OUT		2.4 V	Clear1/Q_1	"	200	"
			88 CKT B	IN	IN	GND	"							"	OUT		2.4 V	Clear1/$\overline{Q}_1$	"	90	"
			89 CKT A				"	IN	IN	2.4 V		OUT	GND	"				Clear2/Q_2	"	200	"
			89 CKT B				"	IN	IN	2.4 V		OUT	GND	"				Clear2/$\overline{Q}_2$	"	90	"
		3003 (Fig. 8)	90 CKT A	IN	J	2.4 V	"							"		OUT	2.4 V	Clock1/Q_1	"	150	"
			90 CKT B	"	"	"	"							"		OUT	"	Clock1/$\overline{Q}_1$	"	70	"
			91 CKT A	"	"	"	"							"	OUT		"	Clock1/Q_1	"	150	"
			91 CKT B	"	"	"	"							"	"		"	Clock1/$\overline{Q}_1$	"	70	"
	t_{PLH}		92 CKT A	"	"	"	"							"	"		"	Clock1/Q_1	"	75	"
			92 CKT B	"	"	"	"							"			"	Clock1/$\overline{Q}_1$	"	50	"
			93 CKT A	"	"	"	"							"		OUT	"	Clock1/Q_1	"	75	"
			93 CKT B	"	"	"	"							"		OUT	"	Clock1/$\overline{Q}_1$	"	50	"
			94 CKT A				"	IN	J	2.4 V		OUT	2.4 V	"				Clock2/Q_2	"	75	"
			94 CKT B				"	"	"	"		OUT	"	"				Clock2/$\overline{Q}_2$	"	50	"
			95 CKT A				"	"	"	"	OUT		"	"				Clock2/Q_2	"	75	"
			95 CKT B				"	"	"	"	"		"	"				Clock2/$\overline{Q}_2$	"	50	"
	t_{PHL}		96 CKT A				"	"	"	"	"		"	"				Clock2/Q_2	"	150	"
			96 CKT B				"	"	"	"	"		"	"				Clock2/$\overline{Q}_2$	"	70	"
			97 CKT A				"	"	"	"		OUT	"	"				Clock2/Q_2	"	150	"
			97 CKT B				"	"	"	"		OUT	"	"				Clock2/$\overline{Q}_2$	"	70	"
10 T_C = +125°C	f_{MAX} 5/	(Fig. 8)	98	IN	B	2.4 V	"							"	OUT		2.4 V	Clock/Q_1	2.5		MHz
			99	IN	B	2.4 V	"							"		OUT	2.4 V	Clock/$\overline{Q}_1$	"		"
			100				"	IN	B	2.4 V		OUT	2.4 V	"				Clock/Q_2	"		"
			101				"	IN	B	2.4 V	OUT		2.4 V	"				Clock/$\overline{Q}_2$	"		"
	t_{PLH}	3003 (Fig. 7)	*102 CKT A	IN	IN	GND	"							"		OUT	2.4 V	Clear1/Q_1	30	125	ns
			*102 CKT B	IN	IN	GND	"							"		OUT	2.4 V	Clear1/$\overline{Q}_1$	"	65	"
			*103 CKT A				"	IN	IN	2.4 V	OUT		GND	"				Clear2/Q_2	"	125	"
			*103 CKT B				"	IN	IN	2.4 V	OUT		GND	"				Clear2/$\overline{Q}_2$	"	65	"
	t_{PHL}		104 CKT A	IN	IN	GND	"							"	OUT		2.4 V	Clear1/Q_1	"	250	"
			104 CKT B	IN	IN	GND	"							"	OUT		2.4 V	Clear1/$\overline{Q}_1$	"	100	"
			105 CKT A				"	IN	IN	2.4 V		OUT	GND	"				Clear2/Q_2	"	250	"
			105 CKT B				"	IN	IN	2.4 V		OUT	GND	"				Clear2/$\overline{Q}_2$	"	100	"
		3003 (Fig. 8)	106 CKT A	IN	J	2.4 V	"							"		OUT	2.4 V	Clock1/Q_1	"	200	"
			106 CKT B	"	"	"	"							"		OUT	"	Clock1/$\overline{Q}_1$	"	85	"
			107 CKT A	"	"	"	"							"	OUT		"	Clock1/Q_1	"	200	"
			107 CKT B	"	"	"	"							"	OUT		"	Clock1/$\overline{Q}_1$	"	85	"

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TABLE III. Group A inspection for device 03 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	Clock 1	Clear 1	K_1	V_{CC}	Clock 2	Clear 2	J_2	$\overline{Q}_2$	Q_2	K_2	GND	Q_1	$\overline{Q}_1$	J_1				
10 T_C = +125°C	t_{PLH}	3003 (Fig. 8)	108 CKT A	IN	J	2.4 V	5.0 V							GND	OUT		2.4 V	Clock 1/Q_1	10	125	ns
			108 CKT B	"	"	"	"							"	OUT		"	Clock 1/Q_1	"	65	"
			109 CKT A	"	"	"	"							"		OUT	"	Clock 1/$\overline{Q}_1$	"	125	"
			109 CKT B				"							"		OUT		Clock 1/$\overline{Q}_1$	"	65	"
			110 CKT A				"	IN	J	2.4 V		OUT	2.4 V	"				Clock 2/Q_2	"	125	"
			110 CKT B				"	"	"	"		OUT	"	"				Clock 2/Q_2	"	65	"
			111 CKT A				"	"	"	"	OUT		"	"				Clock 2/$\overline{Q}_2$	"	125	"
			111 CKT B				"	"	"	"	"		"	"				Clock 2/$\overline{Q}_2$	"	65	"
	t_{PHL}		112 CKT A				"	"	"	"	"		"	"				Clock 2/$\overline{Q}_2$		200	"
			112 CKT B				"	"	"	"	"		"	"				Clock 2/Q_2		85	"
			113 CKT A				"	"	"	"		OUT	"	"				Clock 2/$\overline{Q}_2$		200	"
			113 CKT B				"	"	"	"		OUT	"	"				Clock 2/Q_2		85	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

NOTE: A = normal clock pulse, B = momentary GND, then 4.5 V, E = momentary GND, then open, F = momentary 4.5 V, then GND. J = input pulse $t_p \geq 100$ ns, PRR = 0.5 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Tests shall be performed in sequence.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L $\leq$ 1.5 V when using a high speed checker single comparator.
4/ Input voltages shown are: A = 2.4 V minimum and B = 0.4 V maximum.
5/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
• These tests are performed at device manufacturer's option.
•• Test time limit $\leq$ 100 ns.

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TABLE III. Group A inspection for device 04 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C Test no.	1 1 Clock	2 2 Preset 1	3 3 J_1	4 4 V_{CC}	5 5 Clear	6 6 Preset 2	7 7 K_2	8 8 Q_2	9 9 $\overline{Q}_2$	10 10 J_2	11 11 GND	12 12 $\overline{Q}_1$	13 13 Q_1	14 14 K_1	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{CC}	3005	45 CKT A 45 CKT B 46 CKT A 46 CKT B	GND " " "	4.5 V 4.5 V GND GND	GND " " "	5.5 V " " "	GND GND 4.5 V 4.5 V	4.5 V 4.5 V GND GND	GND " " "			GND " " "	GND " " "			GND " " "	V_{CC} " " "		3.80 2.88 3.80 2.88	mA " " "
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C.																				
7 T_C = +25°C 2/ 4/			47 48 49 50 51 52 53 54 55 56 57	A A B A B A B A B B B	B A " " " A " " " B A	B " " " " " " " " B B	4.5 V " " " " " " " " " "	A A B B B B B B B B B	B A " " " " " " " B A	A " " B " " A A A B B	H 3/ L H L " " H H L H L	L 3/ L H " " " L L H L H	B " " " " " B B B B B	GND " " " " " " " " " "	L 3/ L H " " " L L H L H	H 3/ L H L " " H H L H L	A " B " " " A A A B B	All outputs " " " " " " " " " "	H or L as shown 3/		
8 2/ 4/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C																				
9 T_C = +25°C	f_{MAX} 5/	(Fig. 10)	58 59 60 61	IN " " "	2.4 V 2.4 V	2.4 V 2.4 V	5.0 V " " "	B " " "	2.4 V 2.4 V	2.4 V 2.4 V	OUT	OUT	2.4 V 2.4 V	GND " " "	OUT	OUT	2.4 V 2.4 V	Clock/Q_1 Clock/$\overline{Q}_1$ Clock/Q_2 Clock/$\overline{Q}_2$	3 " " "		MHz " " "
	t_{PLH}	3003 (Fig. 9)	*62 CKT A *62 CKT B *63 CKT A *63 CKT B *64 CKT A *64 CKT B *65 CKT A *65 CKT B	2.4 V " " " " " " "	J J IN IN	2.4 V " " " " " " "	" " " " " " " "	IN IN J J IN IN J J	J J IN IN	2.4 V " " "	OUT OUT	OUT OUT	2.4 V " " "	" " " " " " " "	OUT OUT OUT OUT	OUT OUT	2.4 V " " "	Clear/Q_1 Clear/$\overline{Q}_1$ Preset 1/Q_1 Preset 1/$\overline{Q}_1$ Clear/Q_2 Clear/$\overline{Q}_2$ Preset 2/Q_2 Preset 2/$\overline{Q}_2$	10 " " " " " " "	75 50 75 50 75 50 75 50	ns " " " " " " "
	t_{PHL}		66 CKT A 66 CKT B 67 CKT A 67 CKT B 68 CKT A 68 CKT B 69 CKT A 69 CKT B	GND " " " " " " "	J J IN IN	2.4 V " " " " " " "	" " " " " " " "	IN IN J J IN IN J J	J J IN IN	2.4 V " " "	OUT OUT OUT OUT	OUT OUT	2.4 V " " "	" " " " " " " "	OUT OUT OUT OUT	OUT OUT	2.4 V " " "	Clear/Q_1 Clear/$\overline{Q}_1$ Preset 1/Q_1 Preset 1/$\overline{Q}_1$ Clear/Q_2 Clear/$\overline{Q}_2$ Preset 2/Q_2 Preset 2/$\overline{Q}_2$	10 " " " " " " "	200 90 200 90 200 90 200 90	ns " " " " " " "
	t_{PLH}	3003 (Fig. 10)	70 CKT A 70 CKT B 71 CKT A 71 CKT B 72 CKT A 72 CKT B 73 CKT A 73 CKT B	IN " " " " " " "	2.4 V 2.4 V J J	2.4 V " " " " " " "	" " " " " " " "	J J 4.5 V 4.5 V J J 4.5 V 4.5 V	J J J J	2.4 V " " "	OUT OUT OUT OUT	OUT OUT	2.4 V " " "	" " " " " " " "	OUT OUT OUT OUT	OUT OUT	2.4 V " " "	Clock1/Q_1 Clock1/$\overline{Q}_1$ Clock1/$\overline{Q}_1$ Clock1/$\overline{Q}_1$ Clock2/Q_2 Clock2/$\overline{Q}_2$ Clock2/Q_2 Clock2/$\overline{Q}_2$	10 " " " " " " "	75 50 75 50 75 50 75 50	ns " " " " " " "

See footnotes at end of device type 04.

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Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	Clock	Preset 1	J_1	V_{CC}	Clear	Preset 2	K_2	Q_2	$\overline{Q}_2$	J_2	GND	$\overline{Q}_1$	Q_1	K_1				
9 T_C = +25°C	t_{PHL}	3003 (Fig. 10)	74 CKT A	IN	J	2.4 V	5.0 V	4.5 V						GND		OUT	2.4 V	Clock/Q_1	10	150	ns
			74 CKT B	"	J	"	"	4.5 V						"		OUT	"	Clock/$\overline{Q}_1$	"	70	"
			75 CKT A	"	2.4 V	"	"	J						"	OUT		"	Clock/$\overline{Q}_1$	"	150	"
			75 CKT B	"	2.4 V	"	"	J						"	OUT		"	Clock/$\overline{Q}_1$	"	70	"
			76 CKT A	"			"	4.5 V	J	2.4 V	OUT		2.4 V	"				Clock/Q_2	"	150	"
			76 CKT B	"			"	4.5 V	J	"	OUT		"	"				Clock/$\overline{Q}_2$	"	70	"
			77 CKT A	"			"	J	2.4 V	"		OUT	"	"				Clock/Q_2	"	150	"
			77 CKT B	"			"	J	2.4 V	"		OUT	"	"				Clock/$\overline{Q}_2$	"	70	"
10 T_C = +125°C	f_{MAX} 5/	(Fig. 10)	78	"	2.4 V	2.4 V	"	B						"		OUT	2.4 V	Clock/Q_1	2.5		MHz
			79	"	2.4 V	2.4 V	"	"						"	OUT		2.4 V	Clock/$\overline{Q}_1$	"		"
			80	"			"	"	2.4 V	2.4 V	OUT		2.4 V	"				Clock/Q_2	"		"
			81	"			"	"	2.4 V	2.4 V		OUT	2.4 V	"				Clock/$\overline{Q}_2$	"		"
	t_{PLH}	3003 (Fig. 9)	*82 CKT A	2.4 V	J	2.4 V	"	IN						"	OUT		2.4 V	Clear/$\overline{Q}_1$	10	125	ns
			*82 CKT B	"	J	"	"	IN						"	OUT		"	Clear/Q_1	"	65	"
			*83 CKT A	"	IN	"	"	J						"		OUT	"	Preset1/$\overline{Q}_1$	"	125	"
			*83 CKT B	"	IN	"	"	J						"		OUT	"	Preset1/Q_1	"	65	"
			*84 CKT A	"			"	IN	J	2.4 V		OUT	2.4 V	"				Clear/$\overline{Q}_2$	"	125	"
			*84 CKT B	"			"	IN	J	"		OUT	"	"				Clear/Q_2	"	65	"
			*85 CKT A	"			"	J	IN	"	OUT		"	"				Preset2/$\overline{Q}_2$	"	125	"
			*85 CKT B	"			"	J	IN	"	OUT		"	"				Preset2/Q_2	"	65	"
	t_{PHL}		86 CKT A	GND	J	2.4 V	"	IN						"		OUT	2.4 V	Clear/Q_1	"	250	"
			86 CKT B	"	J	"	"	IN						"		OUT	"	Clear/Q_1	"	300	"
			87 CKT A	"	IN	"	"	J						"	OUT		"	Preset1/$\overline{Q}_1$	"	250	"
			87 CKT B	"	IN	"	"	J						"	OUT		"	Preset1/$\overline{Q}_1$	"	300	"
			88 CKT A	"			"	IN	J	2.4 V	OUT		2.4 V	"				Clock/Q_2	"	250	"
			88 CKT B	"			"	IN	J	"	OUT		"	"				Clock/$\overline{Q}_2$	"	300	"
			89 CKT A	"			"	J	IN	"		OUT	"	"				Preset2/$\overline{Q}_2$	"	250	"
			89 CKT B	"			"	J	IN	"		OUT	"	"				Preset2/$\overline{Q}_2$	"	300	"
	t_{PLH}	3003 (Fig. 10)	90 CKT A	IN	2.4 V	2.4 V	5.0 V	J						GND		OUT	2.4 V	Clock/Q_1	10	125	ns
			90 CKT B	"	2.4 V	"	"	J						"		OUT	"	Clock/$\overline{Q}_1$	"	65	"
			91 CKT A	"	J	"	"	4.5 V						"	OUT		"	Clock/$\overline{Q}_1$	"	125	"
			91 CKT B	"	J	"	"	4.5 V						"	OUT		"	Clock/$\overline{Q}_1$	"	65	"
			92 CKT A	"			"	J	2.4 V	2.4 V	OUT		2.4 V	"				Clock/Q_2	"	125	"
			92 CKT B	"			"	J	2.4 V	"	OUT		"	"				Clock/$\overline{Q}_2$	"	65	"
			93 CKT A	"			"	4.5 V	J	"		OUT	"	"				Clock/Q_2	"	125	"
			93 CKT B	"			"	4.5 V	J	"		OUT	"	"				Clock/$\overline{Q}_2$	"	65	"
	t_{PHL}		94 CKT A	"	J	2.4 V	"	4.5 V						"		OUT	2.4 V	Clock/Q_1		200	"
			94 CKT B	"	J	"	"	4.5 V						"		OUT	"	Clock/$\overline{Q}_1$		85	"
			95 CKT A	"	2.4 V	"	"	J						"	OUT		"	Clock/$\overline{Q}_1$		200	"
			95 CKT B	"	2.4 V	"	"	J						"	OUT		"	Clock/$\overline{Q}_1$		85	"
			96 CKT A	"			"	4.5 V	J	2.4 V	OUT		2.4 V	"				Clock/Q_2		200	"
			96 CKT B	"			"	4.5 V	J	"	OUT		"	"				Clock/$\overline{Q}_2$		85	"
			97 CKT A	"			"	J	2.4 V	"		OUT	"	"				Clock/Q_2		200	"
			97 CKT B	"			"	J	2.4 V	"		OUT	"	"				Clock/$\overline{Q}_2$		85	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

NOTE: A = normal clock pulse, B = momentary GND, then 4.5 V, E = momentary GND, then open, J = input pulse, V_{OL} = 0 V, V_{OH} = 4.5 V.

1/ Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).
2/ Tescs shall be performed in sequence.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V and L ≤ 1.5 V when using a high speed checker single comparator.
4/ Input voltages shown are: A = 2.4 V minimum and B = 0.4 V maximum.
5/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection - device 05 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C / Test no.	1 / 3 / Clock 1	2 / 2 / D_1	3 / 1 / Clear 1	4 / 14 / V_{CC}	5 / 13 / Clear 2	6 / 12 / D_2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q_2	10 / 8 / $\overline{Q}_2$	11 / 7 / GND	12 / 6 / $\overline{Q}_1$	13 / 5 / Q_1	14 / 4 / Preset 1	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C 2/ 3/			47	B	B	B	4.5 V	B	B	B	B	H 4/	H 4/	GND	H 4/	H 4/	B	All outputs	H or L as shown 3/		
			48	-	-	B	-	B	-	-	A	L	-	-	-	-	A	-			
			49	-	-	A	-	A	-	-	A	L	-	-	-	-	A	-			
			50	-	-	-	-	-	-	-	B	H	L	-	L	H	B	-			
			51	A	-	-	-	-	-	A	-	L	L	-	L	H	-	-			
			52	-	B	-	-	B	-	-	-	L	H	-	H	L	-	-			
			53	-	A	-	-	-	A	-	-	L	H	-	H	L	A	-			
			54	-	-	-	-	-	-	-	A	L	L	-	L	H	A	-			
			55	-	-	A	-	A	-	-	A	L	H	-	H	L	B	-			
			56	-	-	-	-	-	-	-	B	H	L	-	L	H	B	-			
			57	-	-	-	-	-	-	-	A	H	H	-	H	L	A	-			
			58	B	-	-	-	-	-	B	-	H	H	-	H	L	-	-			
			59	B	B	-	-	-	-	B	-	H	H	-	H	L	-	-			
			60	A	-	-	-	-	B	A	-	L	H	-	H	L	-	-			
			61	-	-	-	-	-	-	-	B	H	H	-	H	L	B	-			
			62	-	A	B	-	B	-	-	-	H	L	-	L	H	-	-			
			63	-	B	-	-	-	A	-	-	H	L	-	L	H	-	-			
			64	-	-	-	-	-	B	-	A	L	H	-	H	L	A	-			
			65	-	-	A	-	A	-	-	-	L	H	-	H	L	-	-			
			66	B	A	-	-	-	A	-	-	L	H	-	H	L	-	-			
			67	A	-	-	-	-	-	A	-	H	L	-	L	H	B	-			
			68	-	-	-	-	-	-	-	B	H	L	-	L	H	-	-			
			69	-	-	-	-	-	-	-	A	H	L	-	L	H	A	-			
			70	-	-	B	-	B	-	-	-	L	H	-	H	L	-	-			
			71	-	-	A	-	A	-	B	-	L	H	-	H	L	-	-			
			72	-	B	-	-	-	B	A	-	H	L	-	L	H	B	-			
			73	-	B	-	-	-	B	-	A	H	L	-	L	H	A	-			
8 2/ 3/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C																				
9 T_C = +25°C	f_{MAX} 5/	(Fig. 13)	74	IN	IN(H)	5.0 V	5.0 V							GND			B	Clock1/Q_1	3		MHz
			75	IN	IN(G)	5.0 V	"							"	OUT	OUT	B	Clock1/$\overline{Q}_1$	"		"
			76				"	5.0 V	IN(H)	IN	B	OUT		"				Clock2/Q_2	"		"
			77				"	5.0 V	IN(G)	IN	B		OUT	"				Clock2/$\overline{Q}_2$	"		"
	t_{PLH}	3003 (Fig. 11)	78 CKT A			IN	"							"	OUT		J	Clear1/$\overline{Q}_1$	10	75	ns
			78 CKT B			IN	"							"	OUT		J	Clear1/$\overline{Q}_1$	"	65	"
			79 CKT A			J	"							"		OUT	IN	Preset 1/Q_1	"	75	"
			79 CKT B			J	"							"		OUT	IN	Preset 1/Q_1	"	65	"
			80 CKT A				"	IN			J		OUT	"				Clear2/$\overline{Q}_2$	"	75	"
			80 CKT B				"	IN			J		OUT	"				Clear2/$\overline{Q}_2$	"	65	"
			81 CKT A				"	J			IN	OUT		"				Preset 2/Q_2	"	75	"
			81 CKT B				"	J			IN	OUT		"				Preset 2/Q_2	"	65	"
	t_{PHL}		82 CKT A			IN	"							"		OUT	J	Clear/Q_1	"	150	"
			82 CKT B			IN	"							"		OUT	J	Clear/Q_1	"	100	"
			83 CKT A			J	"							"	OUT		IN	Preset 1/$\overline{Q}_1$	"	150	"
			83 CKT B			J	"							"	OUT		IN	Preset 1/$\overline{Q}_1$	"	100	"
			84 CKT A				"	IN			J	OUT		"				Clear/Q_2	"	150	"
			84 CKT B				"	IN			J	OUT		"				Clear/Q_2	"	100	"
			85 CKT A				"	J			IN		OUT	"				Preset 2/$\overline{Q}_{12}$	"	150	"
			85 CKT B				"	J			IN		OUT	"				Preset 2/$\overline{Q}_2$	"	100	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device 05 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				3	2	1	14	13	12	11	10	9	8	7	6	5	4				
				Clock 1	Q_1	Clear 1	V_{CC}	Clear 2	Q_2	Clock 2	Preset 2	Q_2	$\overline{Q}_2$	GND	$\overline{Q}_1$	Q_1	Preset 1				
9 T_C = +25°C	t_{PLH}	3003 (Fig.12)	86 CKT A	IN	IN(G)	B	5.0 V							GND		OUT	5.0 V	Clock1/Q_1	10	100	ns
			86 CKT B	IN	IN(G)	B	•							•		OUT	5.0 V	Clock1/$\overline{Q}_1$	•	65	•
			87 CKT A				•	B	IN(G)	IN	5.0 V	OUT		•				Clock2/Q_2	•	100	•
			87 CKT B				•	B	IN(G)	IN	5.0 V	OUT		•				Clock2/$\overline{Q}_2$	•	65	•
	t_{PHL}		88 CKT A	IN	IN(H)	B	•							•	OUT		5.0 V	Clock1/Q_1	•	150	•
			88 CKT B	IN	IN(H)	B	•							•	OUT		5.0 V	Clock1/$\overline{Q}_1$	•	100	•
			89 CKT A				•	B	IN(H)	IN	5.0 V		OUT	•				Clock2/Q_2	•	150	•
			89 CKT B				•	B	IN(H)	IN	5.0 V		OUT	•				Clock2/$\overline{Q}_2$	•	100	•
	t_{PLH}	3003 (Fig.13)	90 CKT A	IN	IN(G)	5.0 V	•							•	OUT		B	Clock1/Q_1	•	100	•
			90 CKT B	IN	IN(G)	5.0 V	•							•	OUT		B	Clock1/$\overline{Q}_1$	•	65	•
			91 CKT A				•	5.0 V	IN(G)	IN	B		OUT	•				Clock2/Q_2	•	100	•
			91 CKT B				•	5.0 V	IN(G)	IN	B		OUT	•				Clock2/$\overline{Q}_2$	•	65	•
	t_{PHL}		92 CKT A	IN	IN(H)	5.0 V	•							•		OUT	B	Clock1/Q_1	•	150	•
			92 CKT B	IN	IN(H)	5.0 V	•							•		OUT	B	Clock1/$\overline{Q}_1$	•	100	•
			93 CKT A				•	5.0 V	IN(H)	IN	B	OUT		•				Clock2/Q_2	•	150	•
			93 CKT B				•	5.0 V	IN(H)	IN	B	OUT		•				Clock2/$\overline{Q}_2$	•	100	•
10 T_C = +125°C	f_{MAX} 5/	(Fig.12)	94	IN	IN(H)	5.0 V	•							•		OUT	B	Clock1/Q_1	2.5		MHz
			95	IN	IN(G)	5.0 V	•							•	OUT		B	Clock1/$\overline{Q}_1$	•		•
			96				•	5.0 V	IN(H)	IN	B	OUT		•				Clock2/Q_2	•		•
			97				•	5.0 V	IN(G)	IN	B		OUT	•				Clock2/$\overline{Q}_2$	•		•
	t_{PLH}	3003 (Fig.11)	98 CKT A			IN	•							•	OUT		J	Clear1/Q_1	10	125	ns
			98 CKT B			IN	•							•	OUT		J	Clear1/$\overline{Q}_1$	•	85	•
			99 CKT A			J	•							•		OUT	IN	Preset1/Q_1	•	125	•
			99 CKT B			J	•							•		OUT	IN	Preset1/$\overline{Q}_1$	•	85	•
			100 CKT A				•	IN			J		OUT	•				Clear2/Q_2	•	125	•
			100 CKT B				•	IN			J		OUT	•				Clear2/$\overline{Q}_2$	•	85	•
			101 CKT A				•	J			IN	OUT		•				Preset2/Q_1	•	125	•
			101 CKT B				•	J			IN	OUT		•				Preset2/$\overline{Q}_1$	•	85	•

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device 05 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C / Test no.	1 / 3 / Clock 1	2 / 2 / D_1	3 / 1 / Clear 1	4 / 14 / V_{CC}	5 / 13 / Clear 2	6 / 12 / D_2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q_2	10 / 8 / $\overline{Q}_2$	11 / 7 / GND	12 / 6 / $\overline{Q}_1$	13 / 5 / Q_1	14 / 4 / Preset 1	Measured terminal	Test limits Min	Test limits Max	Unit
10 T_C = +125°C	t_{PHL}	3003 (Fig. 11)	102 CKT A			IN	5.0 V							GND		OUT	J	Clear1/Q_1	10	200	ns
			102 CKT B			IN	"							"		OUT	J	Clear1/Q_1	"	120	"
			103 CKT A			J	"							"	OUT		IN	Preset1/$\overline{Q}_1$	"	200	"
			103 CKT B			J	"							"	OUT		IN	Preset1/$\overline{Q}_1$	"	120	"
			104 CKT A				"	IN			J	OUT		"				Clock2/Q_2	"	200	"
			104 CKT B				"	IN			J	OUT		"				Clock2/Q_2	"	120	"
			105 CKT A				"	J			IN		OUT	"				Preset2/$\overline{Q}_2$	"	200	"
			105 CKT B				"	J			IN		OUT	"				Preset2/$\overline{Q}_2$	"	120	"
	t_{PLH}	3003 (Fig. 12)	106 CKT A	IN	IN(G)	B	"							"		OUT	5.0 V	Clock1/Q_1	"	150	"
			106 CKT B	IN	IN(G)	B	"							"		OUT	5.0 V	Clock1/Q_1	"	105	"
			107 CKT A				"	B	IN(G)	IN	5.0 V	OUT		"				Clock2/Q_2	"	150	"
			107 CKT B				"	B	IN(G)	IN	5.0 V	OUT		"				Clock2/Q_2	"	105	"
	t_{PHL}		108 CKT A	IN	IN(H)	B	"							"	OUT		5.0 V	Clock1/$\overline{Q}_1$	"	200	"
			108 CKT B	IN	IN(H)	B	"							"	OUT		5.0 V	Clock1/$\overline{Q}_1$	"	120	"
			109 CKT A				"	B	IN(H)	IN	5.0 V		OUT	"				Clock2/$\overline{Q}_2$	"	200	"
			109 CKT B				"	B	IN(H)	IN	5.0 V		OUT	"				Clock2/$\overline{Q}_2$	"	120	"
	t_{PLH}	3003 (Fig. 13)	110 CKT A	IN	IN(G)	5.0 V	"							"	OUT		B	Clock1/$\overline{Q}_1$	"	150	"
			110 CKT B	IN	IN(G)	5.0 V	"							"	OUT		B	Clock1/$\overline{Q}_1$	"	105	"
			111 CKT A				"	5.0 V	IN(G)	IN	B		OUT	"				Clock2/$\overline{Q}_2$	"	150	"
			111 CKT B				"	5.0 V	IN(G)	IN	B		OUT	"				Clock2/$\overline{Q}_2$	"	105	"
	t_{PHL}		112 CKT A	IN	IN(H)	5.0 V	"							"		OUT	B	Clock1/Q_1	"	200	"
			112 CKT B	IN	IN(H)	5.0 V	"							"		OUT	B	Clock1/Q_1	"	120	"
			113 CKT A				"	5.0 V	IN(H)	IN	B	OUT		"				Clock2/Q_2	"	200	"
			113 CKT B				"	5.0 V	IN(H)	IN	B	OUT		"				Clock2/Q_2	"	120	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

NOTE: A = normal clock pulse, B = momentary GND, then 4.5 V,
J = input pulse $t_p \geq 100$ ns, PRR = 0.5 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Tests shall be performed in sequence.
3/ Input voltages shown are: A = 2.4 V minimum and B = 0.4 V maximum.
4/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L $\leq$ 1.5 V when using a high speed checker single comparator.
5/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for "JAN" marking.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground (zero voltage potential)
V_{IN}	Voltage level at an input terminal
I_{IN}	Current flowing into an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3), and case outline A (see 1.2.3). There is no substitute for case outline C. Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54L71
02	54L72
03	54L73
04	54L78
05	54L74

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6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Manufacturer			
	Circuit A	Circuit B	Circuit C	Circuit D
	Texas Instruments	National Semiconductor		
01		X		
02		X		
03		X		
04		X		
05		X		

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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